INTEGRATED 6-BAR KINEMATIC HARVESTING TOOL WITH GRAPPLING MECHANISM FOR EFFICIENT PEDICLE CUTTING

ABSTRACT

The present disclosure relates to fruits and vegetable harvesting (like cherry tomatoes), via an integrated scissor design interconnected with a spatial or planar mechanism that can hook, pull or both the plant pedicel. The entire kinematic chain is one degree of freedom and does not need multiple modes of actuation. During the operation, when the scissor blade starts to clamp together, the interconnected grappling mechanism hooks the pedicel of the fruit and pulls it towards the cutting area and might lock it in that region until the blades of scissor close and cut the pedicel. Usage of slots, springs, cam profiles, wedges can be done to optimize the cutting and grappling mechanism.

BACKGROUND OF INVENTION

[0001] 1. Field of the Invention

[0002] Manual harvesting of fruits and vegetables is very labor intensive and creates a challenge in Large scale production. Autonomous Harvesting of fruits and veggies have been extensively researched to solve the issues, but still commercial Robots for harvesting intricate yields such as tomatoes still remain a challenge [1]. A major concern is also due to inadequate end effectors which usually are not very accurate.

[0003] Harvesting of delicate yields requires an intricate process of physical manipulation to the target fruit region and extracting either by gripping the fruit itself or cutting the fruit bearing pedicel. [2] The integrated scissor mechanism discussed here is based upon the cutting of the pedicel and is non invasive to the fruit. Autonomous systems that have been deployed for this task usually rely on visual data from cameras. The visual data provides an approximate position of the pedicel and often has discrepancies. This can result in incomplete cuts or damage to the fruit itself. To avoid this, we need a hooking mechanism, best if coupled to the scissor/cutter, which can pull out the pedicel from a zone of possibilities of occurrence as per the detected cutting point, and bring it into the correct region for cutting. Existing end effector focusing on cutting of pedicel, [1, 10-12] lacks the ability of grappling it, thus there is a need for this invention which can perform the aforesaid.

[0004] 2. Prior Arts

[0005] There has been significant advancement in the field of robotic harvesting. Lots of different types of robot end effector have been conceptualized which can grasp, clamp, suck or cut the fruit off the plant and none come close to the design presented by us. In the context of the grasping mechanisms we have end effectors like [3], which contain three finger-like soft projections which surround the fruit, grasp it, and rotate it out of the peduncle. In [4] it contains two finger-like clampers to grip and twist in a similar way. In case of [5] The end-effector was designed with four fingers that have foam sponge pads with motion controlled by solenoid activation. The has also been development of novel flat grippers [6], but the usage of grippers in general do cause damage to the fruit itself and can be challenging specially operating in a cluster of fruits or vegetables. Other studies also displayed various innovative ideas such as the usage of scissor mechanisms coupled together to form a retractable closed cylinder or sphere which can surround the fruit and cut it [7] or the usage of suction force to separate the fruit from its clusters and later cut it [8].

[0006] Various cutting mechanisms have been developed for harvesting of hanging fruits and vegetables but they have been mainly focused on optimizing the cutting by addition of clampers to the main body of the commercially available scissors [9], springs and rubber bearings mounted on scissor with fixed and dynamic fingers [10] or improvement of blade structure and its controlling mechanism with the help of a driving cylinder [11]. Usage of a hook as a support mechanism has also been studied like in the case of [1] where the hook/supporter present at the scissor handle is pulled in and pushed out by the usage of a string and a spring and via kinematic linkages. The supporter itself cannot reach out to hook out any pedicle but is present to aid during cutting.

[0007] The hook mechanism presented in [12] needs multiple modes of actuation and the hooking assembly uses a lot of space around the peduncle, thus making it very difficult to cut out fruits individually in a bunch. Thereafter, Having a specialized hook as presented in which is kinematically coupled to the cutting mechanism we can have major advantages such as pulling out small fruits from a cluster, optimizing the cutting space, avoiding to cause harm to nearby fruits, higher accuracy and precision in cutting.

DESCRIPTION OF THE INVENTION

[0008] Based on the above reasons, the primary objective of the present invention is to provide a mechanism fulfilling the effective cutting of a fruit pedicle during autonomous harvesting. As shown in **Figure 1(a)** and **Figure 1(b)** This Kinematic pedicel cutter provided by the present invention is able to transmit the motion generated by two main arms of the mechanism **(1 & 2)**, housing the blades **(11 & 21)**, to the end of the hooking linkage **(6)**. The whole process is carried out by a single degree of freedom without the requirement of any other modes of actuation for the hook. The main arms of the scissor mechanism can be actuated by an external gripper actuator and its motion and force is transmitted to all the bars of the mechanism. Moreover, as shown in **Figure 1(b),** the mechanism contains the initiation bar **(3)**, supporting bar **(4)**, Handling bar **(5)**. The end point of the hook **(6)** has to follow a predetermined trajectory for the most effective method of harvesting the fruits or vegetables. This method initiates at hooking the fruit pedicle at the right orientation and then pulling it out apart from its bunch. This isolates the fruit pedicle and thus makes it easier to cut out the fruit without causing any harm to the neighboring plant. After cutting the fruit the hook can reopen with the same trajectory in reverse direction to cause minimal disturbance to the plant. This predictive operation of the integrated kinematic cutter leads us to define a definite workspace through which efficient cutting can be provided. Furthermore, the mechanism presented also fulfills the criteria of the appropriate timing and velocity of the hook end point with respect to the movement of the cutter blades so as to perform the entire cutting motion accurately.

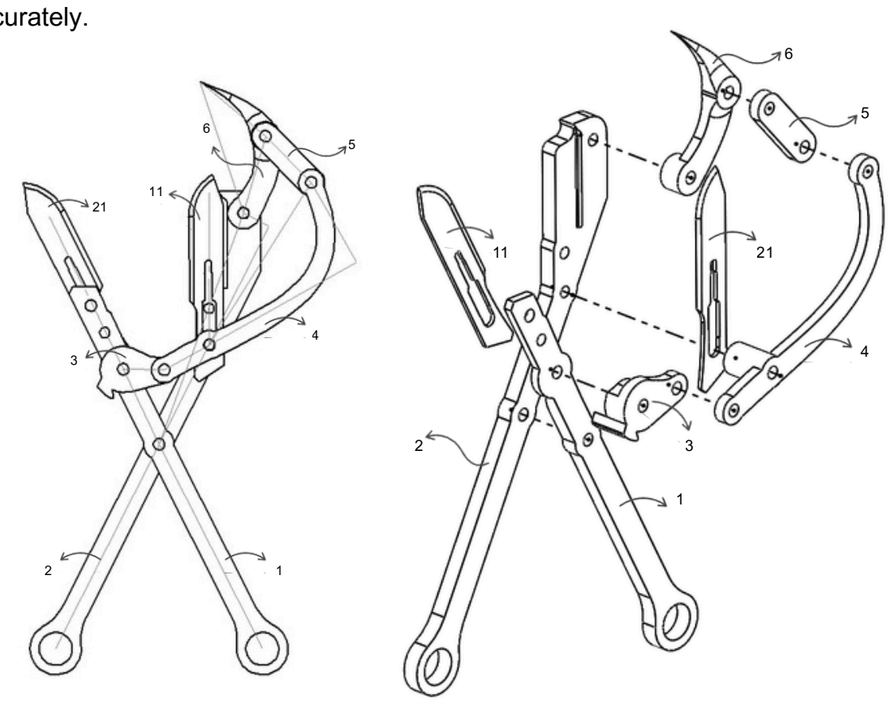


Figure 1: Model of the proposed integrated kinematic cutter with (a) front view in initial position (b) exploded view

WORKING OF THE INTEGRATED KINEMATIC CUTTER

[0009] According to the embodiment of the present invention, to coordinate with the cutting of the blades, the motion of the kinematic mechanism is timed accordingly. The entire motion of hooking and pulling is in sync with the blades reaching the cutting position. The entire motion of the hook can be divided into 3 different time phases with their own specific speeds so as to insure a guaranteed cut. The time phases are i) Hooking phase as shown in between **Figure 2(a)** and **Figure 2(b)**, ii) Pulling phase as shown in between **Figure 2(b)** and **Figure 2(c)**, iii) holding phase as shown in between **Figure 2(b)** and **Figure 2(c)**. During the Hooking phase in the motion of the mechanism, the hook end moves at a very high velocity. This is made to ensure that the pedicle is hooked quickly as soon as the cutting blades begin to actuate towards each other. During this phase the rotational velocity of the hook is much higher in magnitude than the translational

velocity to facilitate proper hooking motion. The next phase of the hook motion is the Pulling phase. It is the longest phase of the entire motion. During this phase the pedicle of the fruit is pulled inside the main cutting area with a good trade-off between the pulling speed and the pulling force. During the end of the pulling phase the speed of the hook further slows down and enters into the holding phase. The cutting of the fruit pedicle is usually observed during the onset of the holding phase. The holding phase is the most stable state of the hook with very little motion. During this phase the pedicle more or less gets locked into the most apt zone of the cutting blades because of the approximately static state of the hook. Thus this entire is observed. The combination of the three phases result in one of the most optimum cutting motion phrasal movement of the hook in correspondence to cutter blades results in highly accurate and efficient cutting.

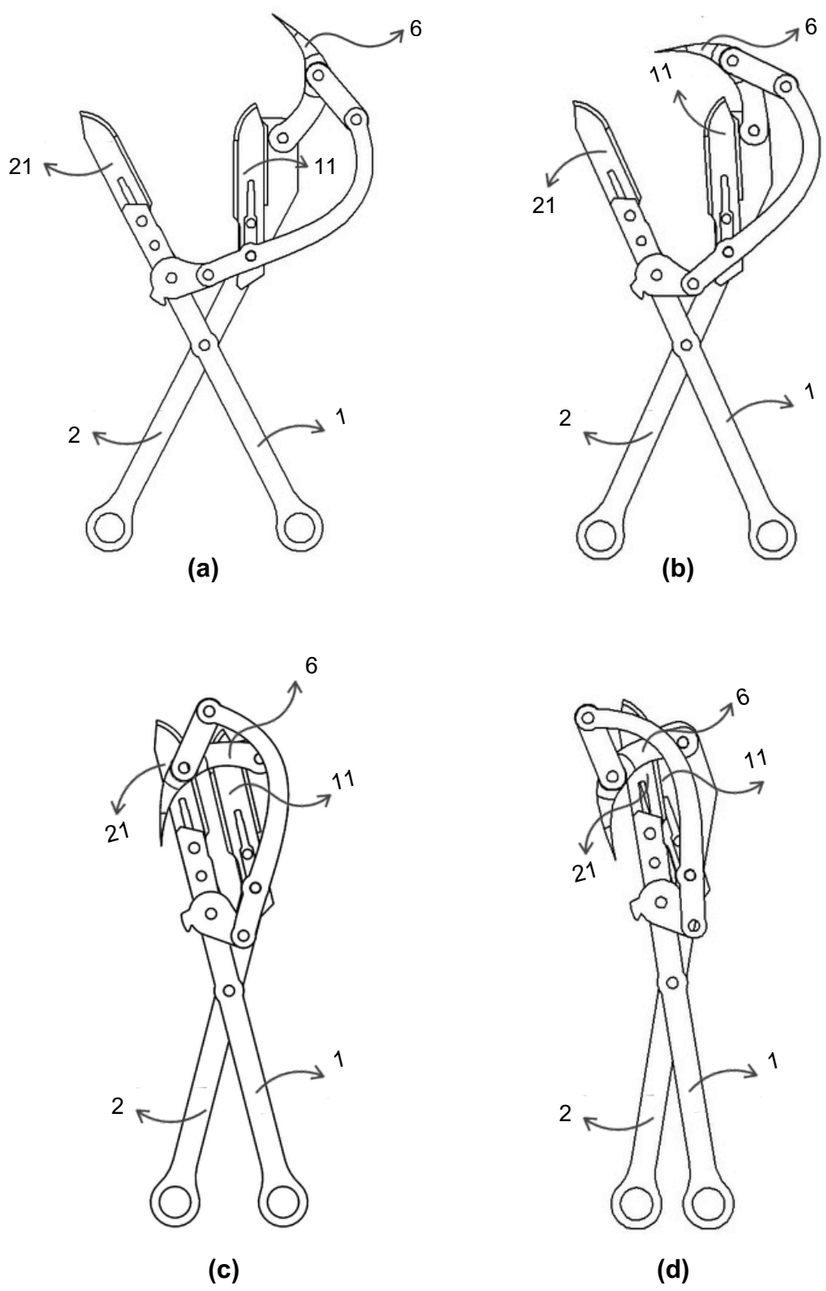


Figure 2: Kinematic motion of the cutter in phases with positions (a) initial position, (b) end of hooking phase, (c) end of pulling phase, (d) end of holding phase.

[0010] According to the embodiment of the present invention, the integrated kinematic cutting mechanism is one degree of freedom and a generally 6 bar mechanism performing the entire harvesting motion. The mechanism can also be an even bar mechanism above a 6 bar mechanism like a 8 bar mechanism or 10 bar mechanism or higher which can perform the same task with higher complexity in motion. With increase in the the number of kinematic linkages of the mechanism, results in an increment of subparts of the end hook. These additional subparts are called the intermediate bars which can form a serial chain ending with the increase in sub-partition of the hook, the hooking motion is achieved by proficient curling of the intermediate bar(s) and the end hook around the fruit pedicel. Incase of 6 bar the curling motion is replaced by curved hook, which results in similar outcome as per higher bar mechanisms where curling is observed. Moreover in an 8 bar or higher cutting kinematic mechanism, the backlash from the fruit pedicle also disturbs the curled hooking shape, thus hindering the pulling force.

[0011] According to the embodiment of the present invention, as per the general case of the 6-bar mechanism, all the bars perform their respective roles to contribute to the final trajectory of the hook. As shown in **Figure 1**. the bars **1** to **6** make up the integrated kinematic mechanism with its kinematic representation

As shown in the kinematic diagram **Figure. 3(b),** where the points A, B, C, O', E, F, H represent the revolute joints of the mechanism, point O is the origin for reference and points L, D, J, G are the essential points on the bars of the mechanism. As seen from front, while closing of the scissor, as the actuation of the bar 1 and bar 2 takes place, the bar 3 (which is also called the initiation bar) immediately rotates clockwise pulling bar 4 (which is also called the supporting bar) downward. The bar 4, pivoted around point B, rotates and pushes the bar 5 (which is also called the handling bar) towards the central cutting region. Since bar 6 or the hook is pivoted to bar 2 at point H, it tends to rotate instantaneously initiating the hooking phase. This phase continues until bar 5 becomes horizontal and the revolute joint points E and F have the same perpendicular distance from origin O. After this the revolute joint point E moves above the point F, thus starting the pulling phase of the mechanism. During this phase the hook starts moving inwards pulling the trapped pedicel into the appropriate cutting spot. This phase continues until bar 5 is inclined few degrees towards the right side from an assumed vertical line drawn at pivot point F. After this phase the holding phase occurs in which bar 5 switches its inclination from right side to its inclination about the left side of the vertical line of pseudo pivot point F. The movement of the hook during this phase is minimal which is crucial to the holding of the hook in the apt cutting region.

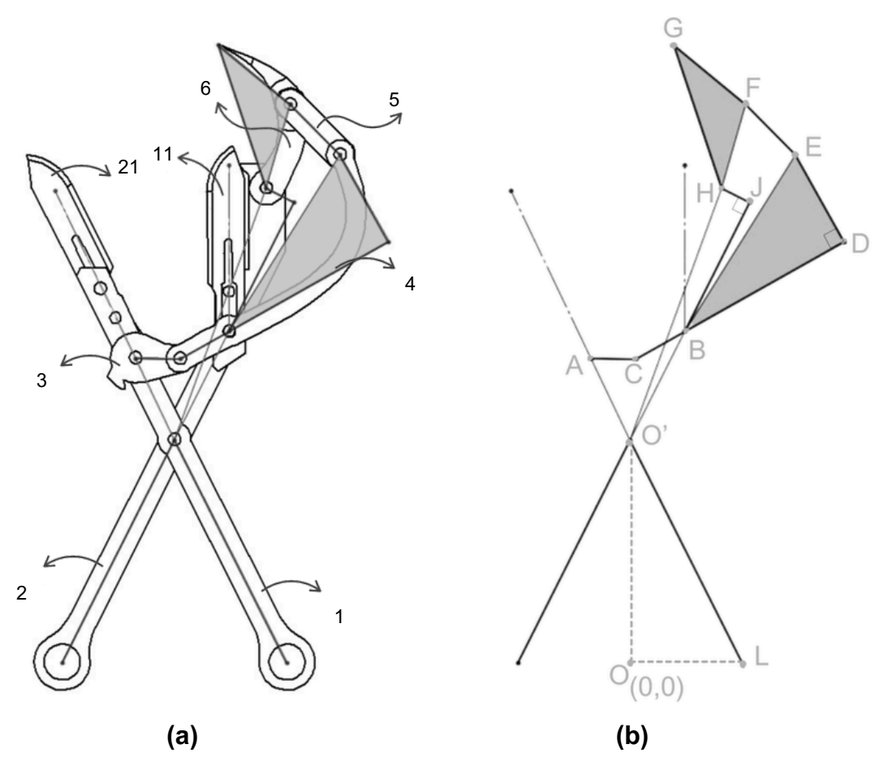
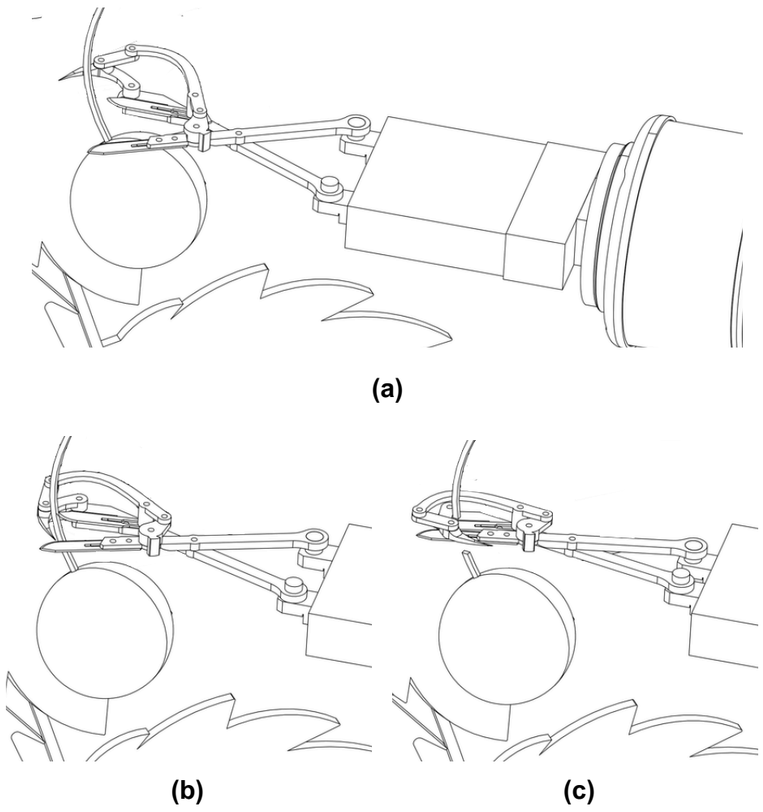


Figure 3: Kinematic structure of the mechanism. (a) The integrated cutter mechanism with kinematic structure. (b) Isolated kinematic structure.

[0012] According to the embodiment of the present invention, As per the previous explanation of the 6 bar version as shown in **Figure 4 (a) - (c)**, the motion of 8 bar and higher cutting mechanism can also be analyzed. In all these cases we assume the facts that, i) we generate the pulling inside motion of the hook end via the handling bar (Bar 5 in case of previously explained 6 bar mechanism) and ii) the hook is linked to n number of intermediate bars (n=1 in case of 8 bar mechanism). iii) The n number of intermediate bars gain their motion via connecting bars pivoted to various links of the mechanism (In case of 8 bar mechanism presented we pivoted the connecting bar to end of the initiation bar {equivalent to bar 3 of the previously explained 6 bar mechanism}). So as explained previously with case 6 bar, similarity here in the case of higher bars, as the two arms of the cutter actuate towards each other, they rotate the initiating bar which in turn pulls down the supporting bar (equivalent to bar 4 of the 6 bar mechanism). The supporting bar transmits its motion to the handling bar, which controls the motion of the hook end. The intermediate bars follow suit, following the trail led by the hook end via the pivots points of the connecting bars. Thus, the hook-end followed by the connecting bars form the basis of the grappling motion of the fruit pedicel. With the increase in the number of bars we can increase the number of time phrases that can be generated and optimize it by adjusting various physical parameters of the bars. This general working principle sums up the entire set of mechanisms that can be generated successfully and thus by tweaking the various parameters such as lengths, size, angles or number of bars, we can generate various sets of cutters suitable for vast operations.



(a)

(b) **(c)**

Figure 4: Cutting of tomato with the invention. (a) Initial position of approach (b) After hooking (c) After pulling and cutting

[0013] According to the embodiment of the present invention, the workspace or the cutting region can be clearly defined. The workspace observed is primarily due to the path followed by the hook end. The kinematic mechanism travelling through various phases of its motion can only cut the pedicle in holding phase, if the pedicle is present in the cutting region initially before the onset of actuation. The kinematic mechanism with the hook end is not symmetrical along its central axis, thus the cutting region is biased towards the mechanism side. Even the blade position of the mechanism side is set in an asymmetrical manner opposing that of the kinematic mechanism. This is done to ensure that the mechanisms do not interfere in cutting during any phase and balanced cutting is observed. For the case of 6 bar kinematic cutting mechanism, the hook's trajectory (x,y) can be easily defined with respect to the input angle θ, which is the angle observed between the two main arms of the cutter when actuated. All angles are assumed to be in radian and the lengths in millimeter. Thus the value of trajectory point (x,y) with respect to input angle θ is -

$$x = \frac{l_2 \sin(\theta/2 - \lambda)}{\cos\lambda} - d\cos\alpha \quad \& \quad y = l_1 \cos(\theta/2) + \frac{l_2 \cos(\theta/2 - \lambda)}{\cos\lambda} + d\sin\alpha$$

where -

$$\alpha = \pi - \psi_2 + \theta/2 + \cos^{-1}\left(\frac{(l_2\tan\lambda)^2 + b_3^2 - b_2^2}{2 * l_2\tan\lambda * b_3}\right) - \cos^{-1}\left(\frac{a_6^2 + b_3^2 - a_7^2}{2a_6 b_3}\right)$$

$$b_3 = \sqrt{(l_2\tan\lambda)^2 + b_2^2 - 2 * l_2\tan\lambda * b_2\cos\gamma_5}$$

$$b_2 = \sqrt{a_5^2 + l_3^2 - 2a_5 l_3\cos\Phi_4}$$

$$\gamma_5 = \pi/2 + \sin^{-1}(a_5\sin\Phi_4/b_2)$$

$$\Phi_4 = \sin^{-1}(a_1\sin\theta/b_1) - \cos^{-1}\left(\frac{a_4^2 + b_1^2 - a_3^2}{2a_4 b_1}\right) - \psi_1$$

$$b_1 = \sqrt{a_1^2 + a_2^2 - 2a_1 a_2\cos\theta}$$

where angle α is the output angle and the other variables such as a_1, a_2, a_5, a_4, a_6, a_7, b_2, b_3, λ, l_1, l_2, $\angle\psi_2$, $\angle\psi_1$, d are the respective lengths and angles shown in **Figures 5 - 7.** The above set of equations are fixed constants determining the geometry of the kinematic cutting mechanism

[0014] According to the embodiment of the present invention, the mechanism is kinematically constrained. It means that the mechanism cannot open or close more than the set limit by the dimension of its bars. This is essential as to prevent any overshoot of linkages. Its it specially essential in the case of return stroke when the hook takes its initial position with high velocity. If the mechanism is not constrained kinematically then in the aforesaid scenario the hook might cross its desired location because of its inertia and thus a singularity is observed. To prevent this singularity, the initiation bar and the supporting bar are coupled in a manner such that during the maximum desired extension their central axis align and any further actuation would violate the geometry. Moreover, the revolute joint between the initiation bar and the supporting bar is not kept on the central axis, but rather kept in the side opposite to the hook side, so as to constraint the mechanism before the hook can overshoot. In case of the 6 bar **k**inematic cutting mechanism with a specific case of dimension shown in the **Figure 4**, the value of maximum input angle θ is constrained at 55.61 degrees and can vary according to the chosen dimensions of the geometries.

[0015] According to the embodiment of the present invention, during the final holding phase, the position of the hook is maintained at an approximating static state so as to lock the pedicel in the most appropriate position for cutting. This state is achieved because of the handling bar which tries to keep the vertical position of revolute joint point F from the origin O, almost constant during the holding phase. If we consider a vertical axis from point F called handling axis, during the onset of the holding phase, the handling bar is inclined about 15 degrees to the right side of the handling axis, as seen from front view. After this any actuation provided to the mechanism will result primarily in the rotation of the holding bar about point F which is approximately fixated. Thus only the inclination of the holding bar changes from right to left side of the holding axis. This continues until the hook inclination has switched it direction to the left side of the holding axis, with a maximum inclination of 10 degrees. After this the mechanism geometrically constrained as it reached its maximum closing state.

[0016] According to the embodiment of the present invention, the kinematic cutting mechanism has to be properly oriented when it is used for harvesting. This orientation must be facilitated by use of various robotic systems, automotives, or machineries. The invention presented here, is usually actuated by a set of industrial grippers who opens and closes the main arms of the mechanism thus providing appropriate velocity and force to the system. The gripper can be mounted on industrial robots such as a serial arm which can reach and orient the cutter to best alignment for cutting the pedicels. The arm can be mounted on any vehicle/rail that travel to the various parts of the farm. We can also implement various detection and approaching mechanisms that can assist in reaching the pedicel. One such architecture is is presented in the flow chart **Fig. 13**, where we use point cloud data and control systems to harvest the fruits. According to the embodiment of the present invention, the invention can also have cam slots so as to ensure the smooth operation of the revolute joints without any misalignment. The overlapping bars of the mechanism need to be placed with clearances so as to prevent any collision during operation. All the learances also need to account for the proximity of the cutting blades. Usage of thinner bars of in mechanism could help in bringing the hook closer to the cutting blades and thus increasing its capability to hook smaller pedicles. But using thinner bars would compromise with the strength and pulling force of the mechanism. Usage of bearing in some of the revolute joints can facilitate smooth rotation and handle unexpected loads. Usage of compression / extension springs can boost the overall robustness of the system. The blade and hook geometry also play a very crucial role in determining the outcome of the harvest.

DERIVATION OF THE HOOK TRAJECTORY OF THE INVENTION

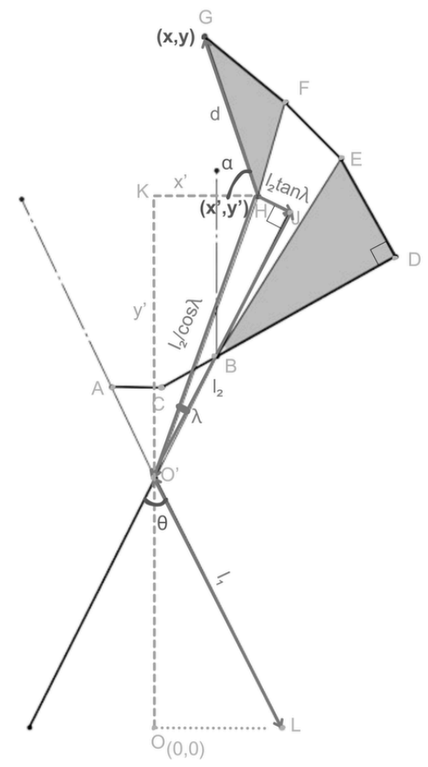


Figure 5: Kinematic structure of the entire mechanism of cutter

[0017] To determine the region where we can place the pedicel of fruit to be harvested, we must know the trajectory the hook end follow during the entire operation of closing the kinematic mechanism. Thus to derive the governing equation for the trajectory of the hook end (x,y) with respect to input angle θ and scissor constants in the 6 bar kinematic mechanism, we define parameters such as

[0018] As shown in **Figure 5 to Figure 7**,

Origin point :- O

Revolute joint points :- A, B, C, O', E, F, H

Geometry points on bars :- L, D, J

Hook trajectory points :- (x,y) {Hook end point G} and (x',y') {Pivot point H}

Input angle :- θ

Output angle :- α

Scissor constants (They do not change in operation) :- Lengths - l_1, l_2, l_3, d, a_1, a_2, a_3, a_4, a_5, a_6, a_7

Scissor constants (They do not change in operation) :- Angles - λ, ψ_1, ψ_2, ψ_3

Variables lengths during operation :- b_1, b_2, b_3

Variables angles operation :- Φ_1, Φ_2, Φ_3, Φ_4, Φ_5, Φ_6, Φ_7, γ_1, γ_2, γ_3, γ_4, γ_5

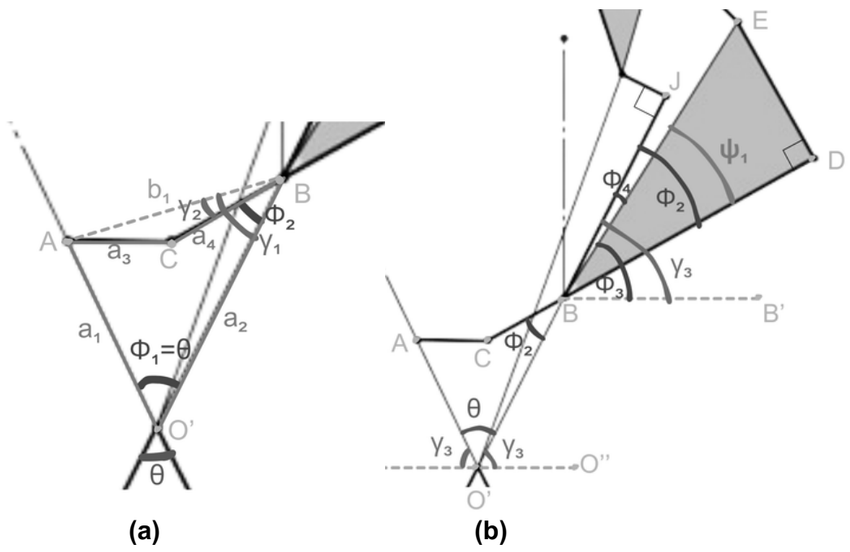


Figure 6: Kinematic structure of a portion of mechanism (a) central region (b) central and skewed toward right.

[0019] Now, according to **Fig. 5**, In △*GHK*

$x = x' - d\cos\alpha$
$y = y' + d\sin\alpha$ } *eq. 1*

Now, *KH* = x', *OK* = y', ∠*KO'H* = θ/2 - λ

$$\Rightarrow x' = \frac{l_2 \sin(\theta/2 - \lambda)}{\cos\lambda} \;\&\; y' = l_1\cos(\theta/2) + \frac{l_2\cos(\theta/2 - \lambda)}{\cos\lambda}$$ } *eq. 2*

By *eq.1* and *eq. 2* we get :-

$$x = \frac{l_2 \sin(\theta/2 - \lambda)}{\cos\lambda} - d\cos\alpha \quad \& \quad y = l_1\cos(\theta/2) + \frac{l_2\cos(\theta/2 - \lambda)}{\cos\lambda} + d\sin\alpha$$ } *eq. 3*

[0020] From **Figure 6 (a)**, $\Phi_1 = \theta$

In △*AO'B,* by cosine law -

$b_1 = \sqrt{a_1^2 + a_2^2 - 2a_1a_2\cos\theta}$ } *eq. 4*

By sine law -

$a_1/\sin\gamma_1 = b_1/\sin\theta$

$\Rightarrow \gamma_1 = \sin^{-1}(a_1\sin\theta/b_1)$ } *eq. 5*

In △*ABC*, by cosine law -

$$\gamma_2 = \cos^{-1}\left(\frac{a_4^2 + b_1^2 - a_3^2}{2a_4b_1}\right)$$ } *eq. 6*

∵ $\Phi_2 = \gamma_1 - \gamma_2$, ∴ by *eq. 4, 5* & *6*

$$\Phi_2 = \sin^{-1}(a_1\sin\theta/b_1) - \cos^{-1}\left(\frac{a_4^2 + b_1^2 - a_3^2}{2a_4b_1}\right)$$ } *eq. 7*

where, $b_1 = \sqrt{a_1^2 + a_2^2 - 2a_1a_2\cos\theta}$

[0021] From **Figure 6 (b)**,

$\gamma_3 = \pi/2 - \theta/2$ } *eq. 8*

$\Phi_4 = \Phi_2 - \psi_1$ } *eq. 9*

From *eq. 9* and *eq. 7*

$$\Phi_4 = \sin^{-1}(a_1\sin\theta/b_1) - \cos^{-1}\left(\frac{a_4^2 + b_1^2 - a_3^2}{2a_4b_1}\right) - \psi_1$$ } *eq. 10*

$\Phi_3 = \gamma_3 - \Phi_4$; thus by *eq. 8* and *eq. 10*

$$\Rightarrow \Phi_3 = \pi/2 - \theta/2 + \psi_1 - \sin^{-1}(a_1\sin\theta/b_1) + \cos^{-1}\left(\frac{a_4^2 + b_1^2 - a_3^2}{2a_4b_1}\right)$$ } *eq. 11*

[0021] From **Figure 7 (a) and (b)**,

$\Phi_7 = \gamma_3 + \lambda$; thus by eq. 8

$\Phi_7 = \pi/2 - \theta/2 + \lambda$ } eq. 12

Around point H,

$2\pi = \alpha + \psi_2 + \Phi_5 + \Phi_6 + \Phi_7 + \psi_3 \Rightarrow$

$\alpha = 2\pi - (\psi_2 + \psi_3 + \Phi_5 + \Phi_6 + \Phi_7)$ } eq. 13

In △HJO',

$\psi_3 = \pi/2 - \lambda$ } eq. 14

In △BJE, by cosine rule,

$b_2 = \sqrt{a_5^2 + l_3^2 - 2a_5l_3\cos\Phi_4}$ } eq. 15

by sine rule,

$a_5/\sin(\pi - \gamma_4) = b_2/\sin\Phi_4 \Rightarrow$

$\gamma_4 = \pi - \sin^{-1}(a_5\sin\Phi_4/b_2)$ } eq. 16

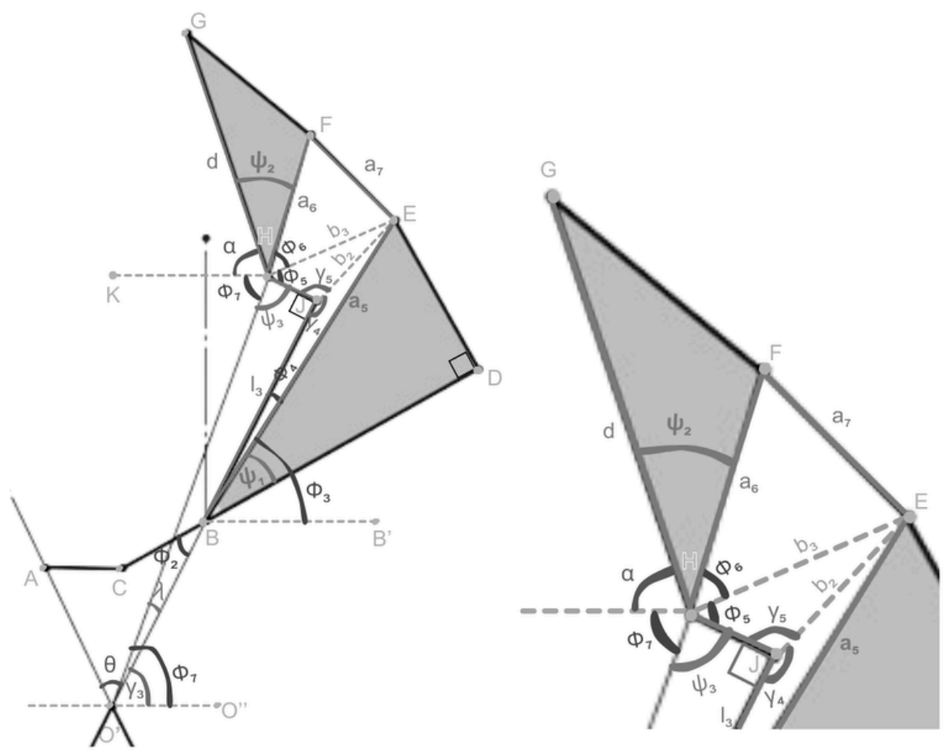


Figure 7: Kinematic structure of a portion of mechanism (a) right region (b) top right region.

Around point *J*,

$2\pi = \gamma_4 + \gamma_5 + \pi/2 \Rightarrow$ by *eq. 16*

$\gamma_5 = \pi/2 + \sin^{-1}(a_5\sin\Phi_4/b_2)$ } *eq. 17*

In △*HJE*

$HJ = l_2\tan\lambda$ *(from* **Figure 5***)* ,

by cosine law

$b_3 = \sqrt{(l_2\tan\lambda)^2 + b_2^2 - 2{*}l_2\tan\lambda{*}b_2\cos\gamma_5}$ } *eq. 18*

Now by cosine law again,

$$\Phi_5 = \cos^{-1}\left(\frac{(l_2\tan\lambda)^2 + b_3^2 - b_2^2}{2{*}l_2\tan\lambda{*}b_3}\right)$$ } *eq. 19*

In △*HFE*, by cosine rule

$$\Phi_6 = \cos^{-1}\left(\frac{a_6^2 + b_3^2 - a_7^2}{2a_6\,b_3}\right)$$ } *eq. 20*

∴ In *eq. 13*,

$\alpha = 2\pi - \psi_2 - \psi_3 - \Phi_5 - \Phi_6 - \Phi_7$ put *eq. 12, 14, 19, 20* =>

$$\alpha = 2\pi - \psi_2 - \cos^{-1}\left(\frac{(l_2\tan\lambda)^2 + b_3^2 - b_2^2}{2{*}l_2\tan\lambda{*}b_3}\right) - \cos^{-1}\left(\frac{a_6^2 + b_3^2 - a_7^2}{2a_6\,b_3}\right) - \pi/2 + \theta/2 - \lambda - \pi/2 + \lambda$$ } *eq. 21*

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[0022] Thus by equations 3, 4, 10, 15, 17, 18, 21 we can finally state the hook trajectory (x,y) with respect to input angle θ as -

$$x = \frac{l_2 \sin(\theta/2 - \lambda)}{\cos\lambda} - d\cos\alpha \quad \& \quad y = l_1\cos(\theta/2) + \frac{l_2\cos(\theta/2 - \lambda)}{\cos\lambda} + d\sin\alpha$$

where :-

$$\alpha = \pi - \psi_2 + \theta/2 + \cos^{-1}\left(\frac{(l_2\tan\lambda)^2 + b_3^2 - b_2^2}{2{*}l_2\tan\lambda{*}b_3}\right) - \cos^{-1}\left(\frac{a_6^2 + b_3^2 - a_7^2}{2a_6\,b_3}\right)$$

$b_3 = \sqrt{(l_2\tan\lambda)^2 + b_2^2 - 2{*}l_2\tan\lambda{*}b_2\cos\gamma_5}$

$b_2 = \sqrt{a_5^2 + l_3^2 - 2a_5l_3\cos\Phi_4}$

$\gamma_5 = \pi/2 + \sin^{-1}(a_5\sin\Phi_4/b_2)$

$$\Phi_4 = \sin^{-1}(a_1\sin\theta/b_1) - \cos^{-1}\left(\frac{a_4^2 + b_1^2 - a_3^2}{2a_4b_1}\right) - \psi_1$$

$b_1 = \sqrt{a_1^2 + a_2^2 - 2a_1a_2\cos\theta}$

RESULT & CONCLUSION

[0023] Thus the governing equation of the hook has been derived using trigonometric identities to eliminate the 20 unknowns with assuming the respective mechanism structure. The final kinematic cutting mechanism can be used in cases of precise cutting and eliminating any discrepancies during cutting. The entire motion is done in 3 phases viz. hooking phase, pulling phase and holding phase. Thus the final model can be 3D printed and tested in real world with tomatoes as shown in **Figure 8.** The design can further be improved by



Figure 8: 3-D printed prototype of the kinematic cutting mechanism

using bearings and spring. Overall the performance is good and we can plot the governing equation in MATLAB to see the actual hook trajectory to predict the accurate cutting region where the pedicle of fruit like tomato, can be present. The governing from **Figure 5 to Figure 7** is given as follows -

$$x = \frac{l_2 \sin(\theta/2 - \lambda)}{\cos\lambda} - d\cos\alpha \quad \& \quad y = l_1\cos(\theta/2) + \frac{l_2\cos(\theta/2 - \lambda)}{\cos\lambda} + d\sin\alpha$$

where :-

$$\alpha = \pi - \psi_2 + \theta/2 + \cos^{-1}\left(\frac{(l_2\tan\lambda)^2 + b_3^2 - b_2^2}{2*l_2\tan\lambda*b_3}\right) - \cos^{-1}\left(\frac{a_6^2 + b_3^2 - a_7^2}{2a_6 b_3}\right)$$

$$b_3 = \sqrt{(l_2\tan\lambda)^2 + b_2^2 - 2*l_2\tan\lambda*b_2\cos\gamma_5}$$

$$b_2 = \sqrt{a_5^2 + l_3^2 - 2a_5 l_3\cos\Phi_4}$$

$$\gamma_5 = \pi/2 + \sin^{-1}(a_5\sin\Phi_4/b_2)$$

$$\Phi_4 = \sin^{-1}(a_1\sin\theta/b_1) - \cos^{-1}\left(\frac{a_4^2 + b_1^2 - a_3^2}{2a_4 b_1}\right) - \psi_1$$

$$b_1 = \sqrt{a_1^2 + a_2^2 - 2a_1 a_2\cos\theta}$$

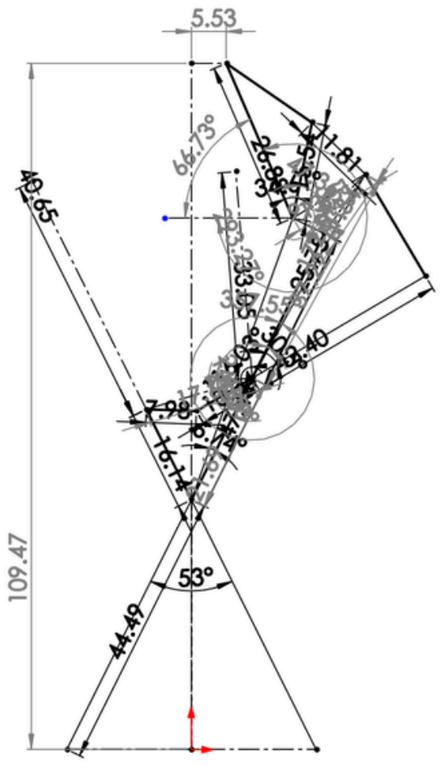


Figure 9: Kinematically constraint Sketch made in SolidWorks

[0024] The values of the constants of the governing equations according to the prototype shown in **Figure 8** are (all lengths are in **mm**) -

a_1 = 16.1355, a_2 = 21.6918, a_3 = 7.9799, a_5 = 37.2152, a_6 = 15.5387, a_4 = 10.3386, a_7 = 11.8131, $\angle\lambda$ = 6.7407°, l_1 = 44.4941, l_2 = 47.4422, $\angle\psi_2$ = 34.6088°, $\angle\psi_1$ = 30.4996, d = 26.88.

[0025] All these values can be used to make a kinematically constraint sketch in SolidWorks as shown in **Figure 9** in order to verify the governing equation at various input points such as **θ = 53°**, **θ = 38°** and **θ = 23°.** Thus the value of output angle α and end point (x, y) with respect to origin O shown in **Figure 5**, can be compared between the result obtained by kinematically constraint sketch of SolidWorks and result by our derived governing equation as -

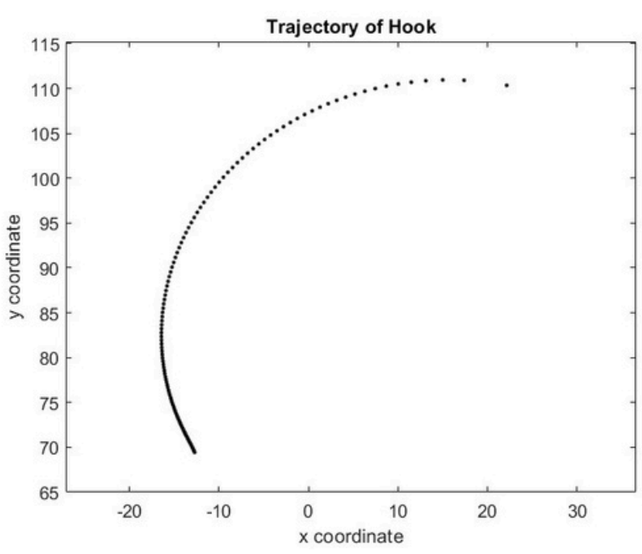
For Input angle θ = 53°	output angle α	Hook end x coordinate	Hook end y coordinate
Result by kinematically constraint sketch in SolidWorks	66.7358	5.5334	109.4732
Result by the derived governing equation plotted in MATLAB	66.7359	5.5336	109.4734

For Input angle θ = 38°	output angle α	Hook end x coordinate	Hook end y coordinate
Result by kinematically constraint sketch in SolidWorks	-9.6729	-16.3539	84.2363
Result by the derived governing equation plotted in MATLAB	-9.6731	-16.3540	84.2365

For Input angle θ = 23°	output angle α	Hook end x coordinate	Hook end y coordinate
Result by kinematically constraint sketch in SolidWorks	-50.0639	-13.2914	70.5978
Result by the derived governing equation plotted in MATLAB	-50.0640	-13.2915	70.5980

[0026] Thus the derived governing equation accurately predicts all the values that are obtained by the SolidWorks computational model. The miniscule error seen the tables is only round-off error and the governing equation fully accurate. Thus we can plot the governing equation on MATLAB and visualize the hook end's trajectory in **Figure 10(a)**. We can also see the variation of output angle (α) with respect to the input angle (θ) in **Figure 10(b)**.

(a)



(b)

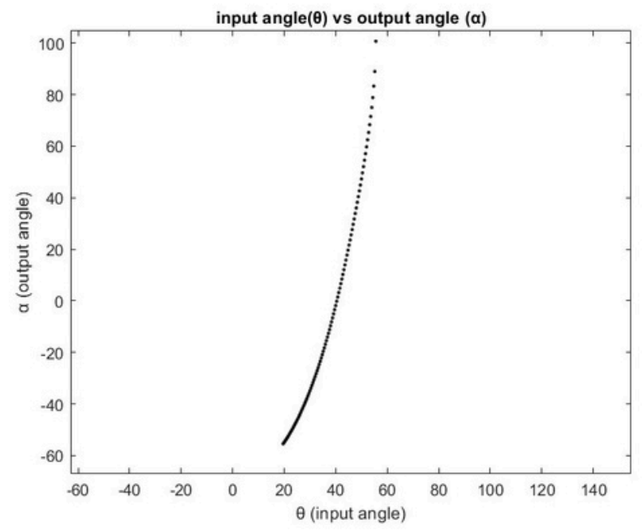


Figure 10: Graphs obtained by plotting the governing equation in MATLAB (a) Trajectory of hook end (x, y) and (b) variation of output angle (α) to input angle (θ)

[0026] We can see the various phases of hook movement are clearly represented in **Figure 10(a).** Initial actuation of the the cutter results in the hooking phase, where the hooks rotates very fast, which is clearly shown as points are far apart. In the next pulling the hook pulls the pedicel inside to cut which is clearly represented. In the last holding phase the speed decreases and the hook position is almost static, which is shown by dark region in the end as the points are very close to each other.

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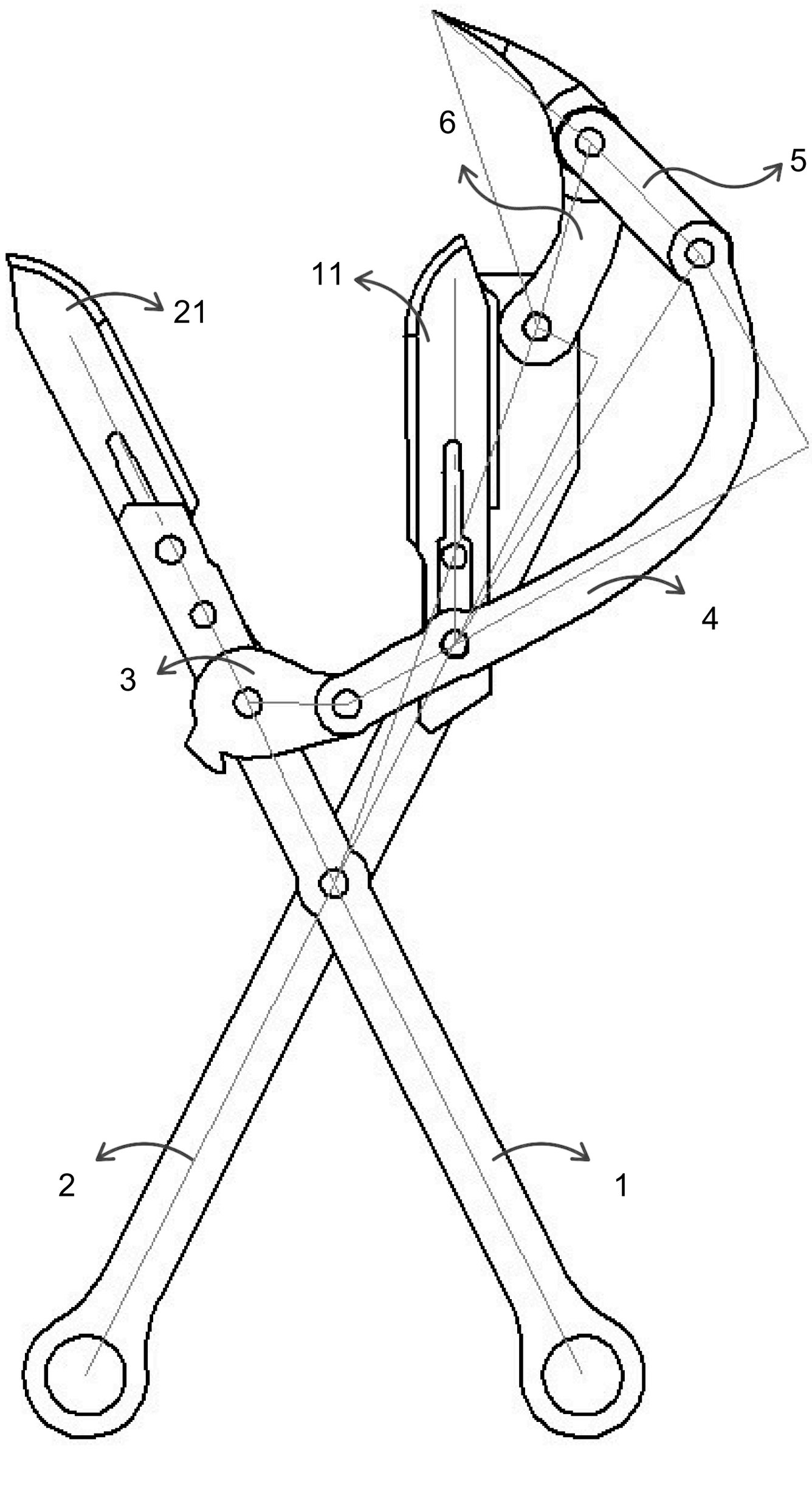


Fig 1. (a)

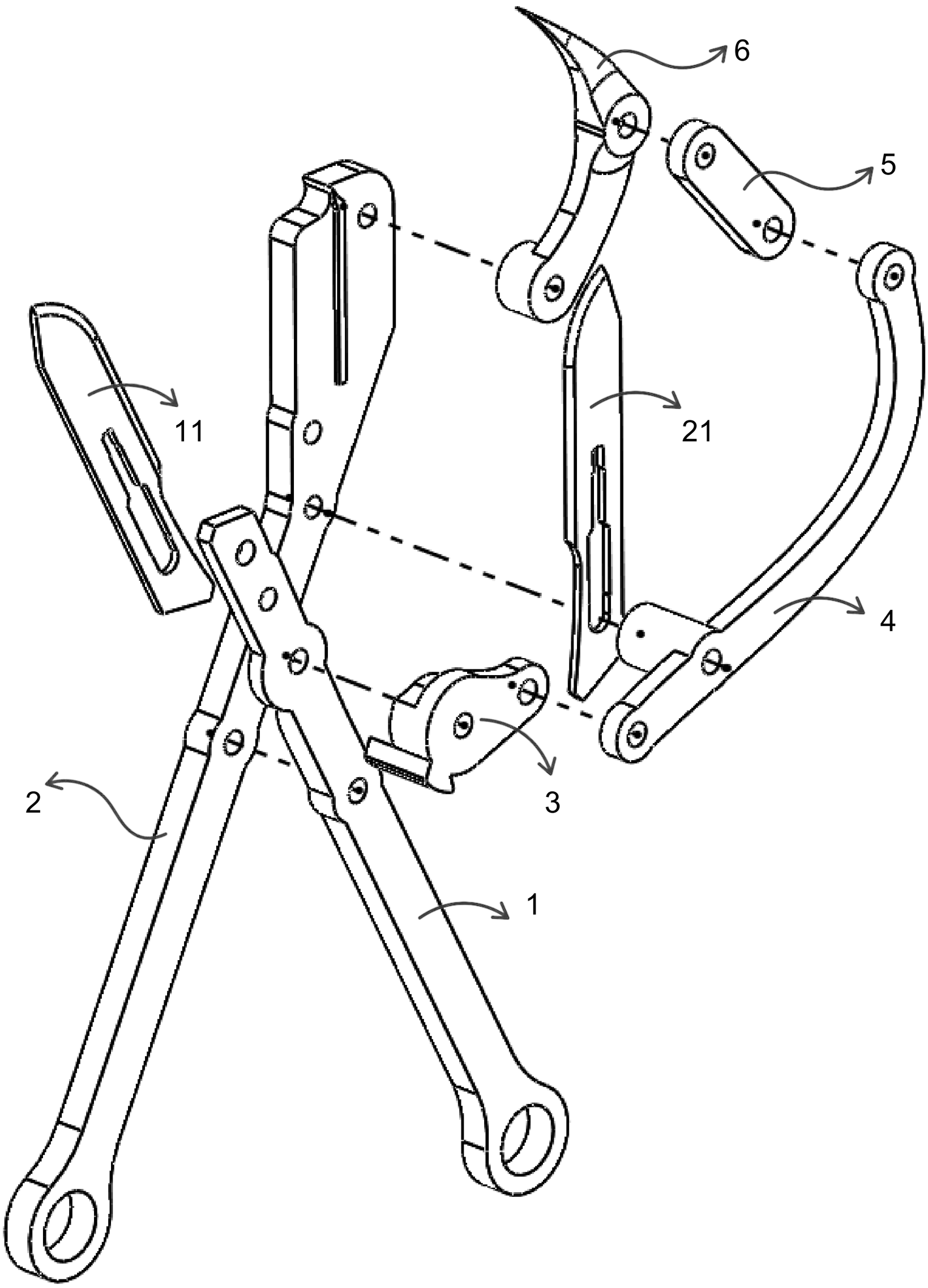


Fig 1. (b)

Fig 2.

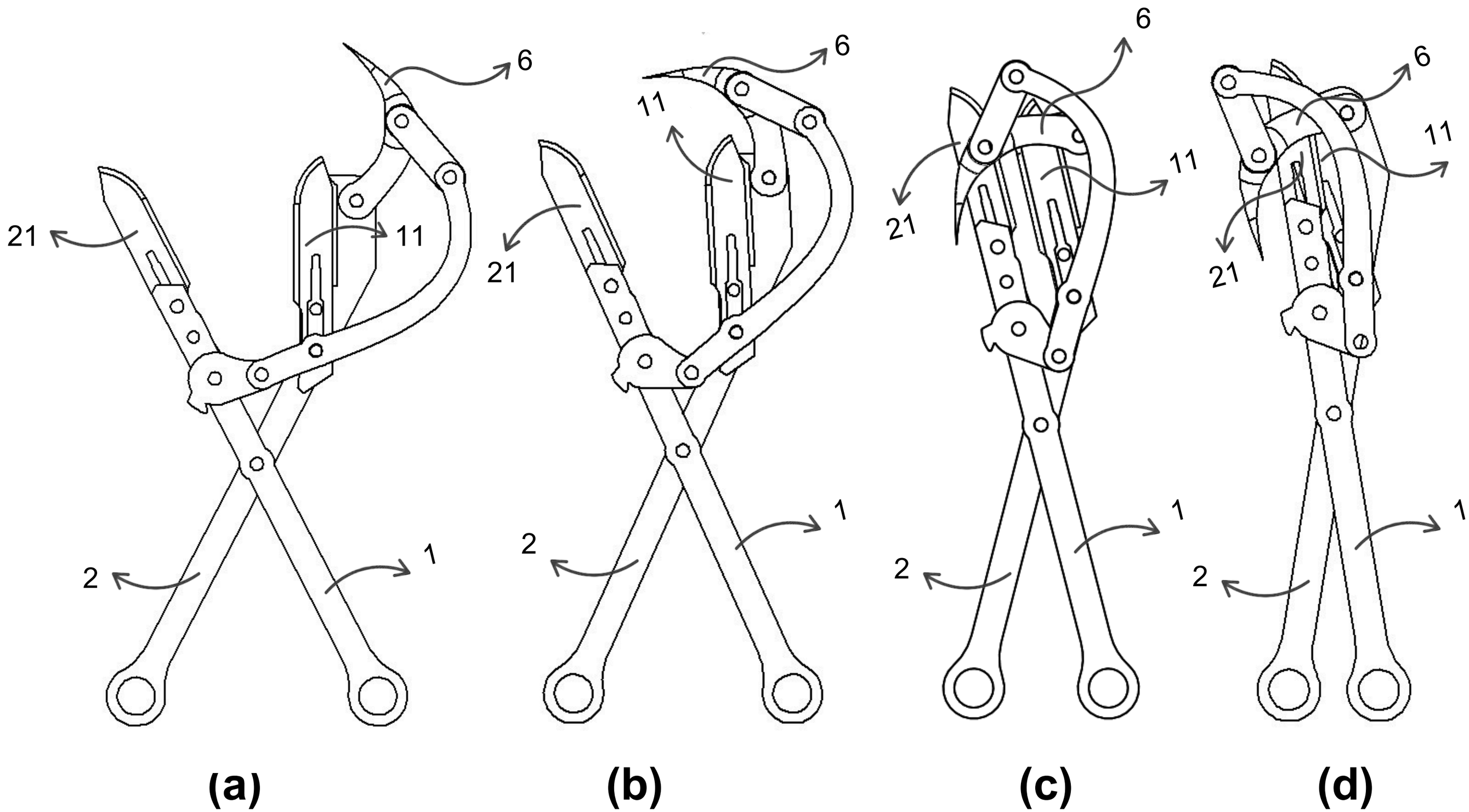
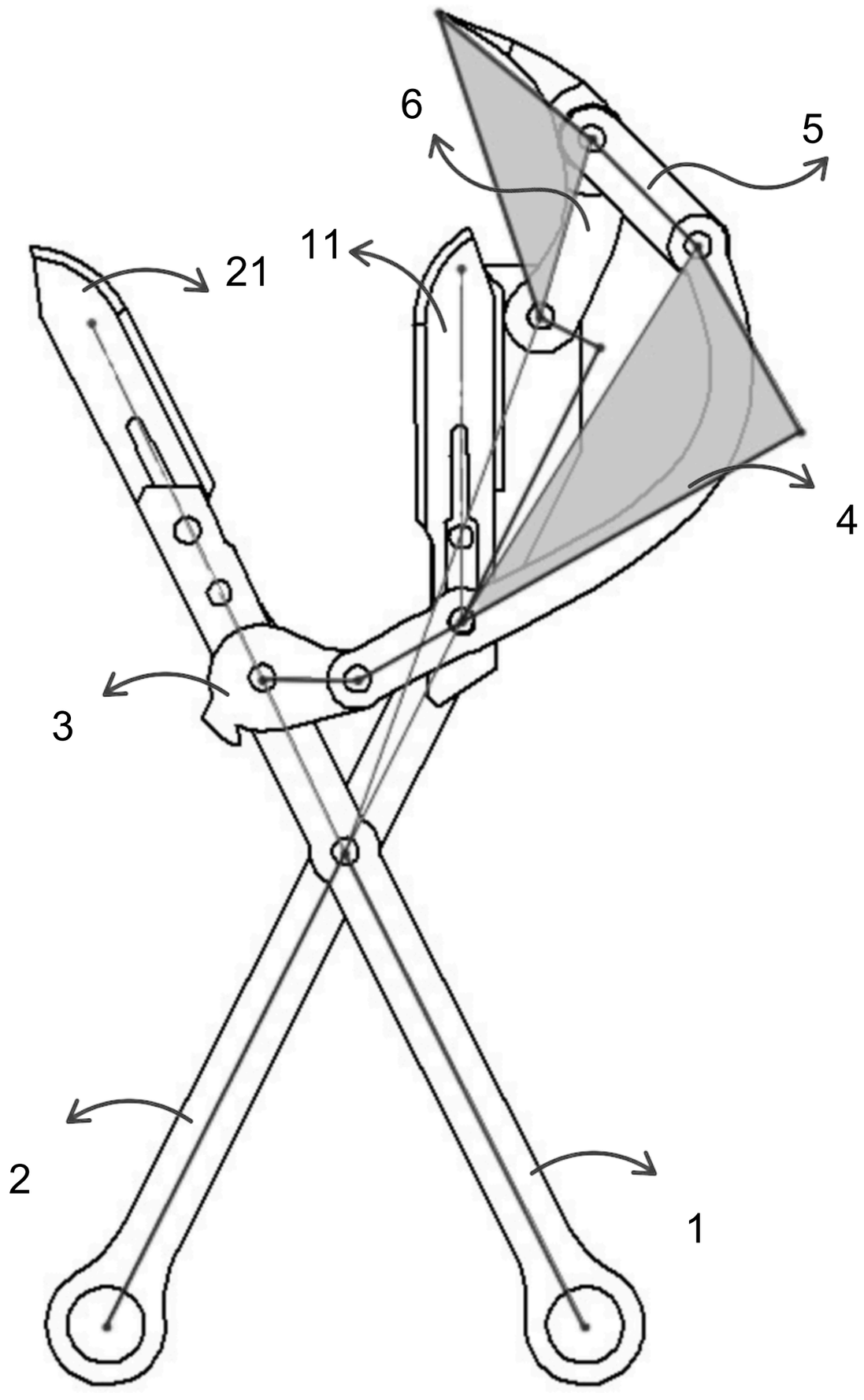
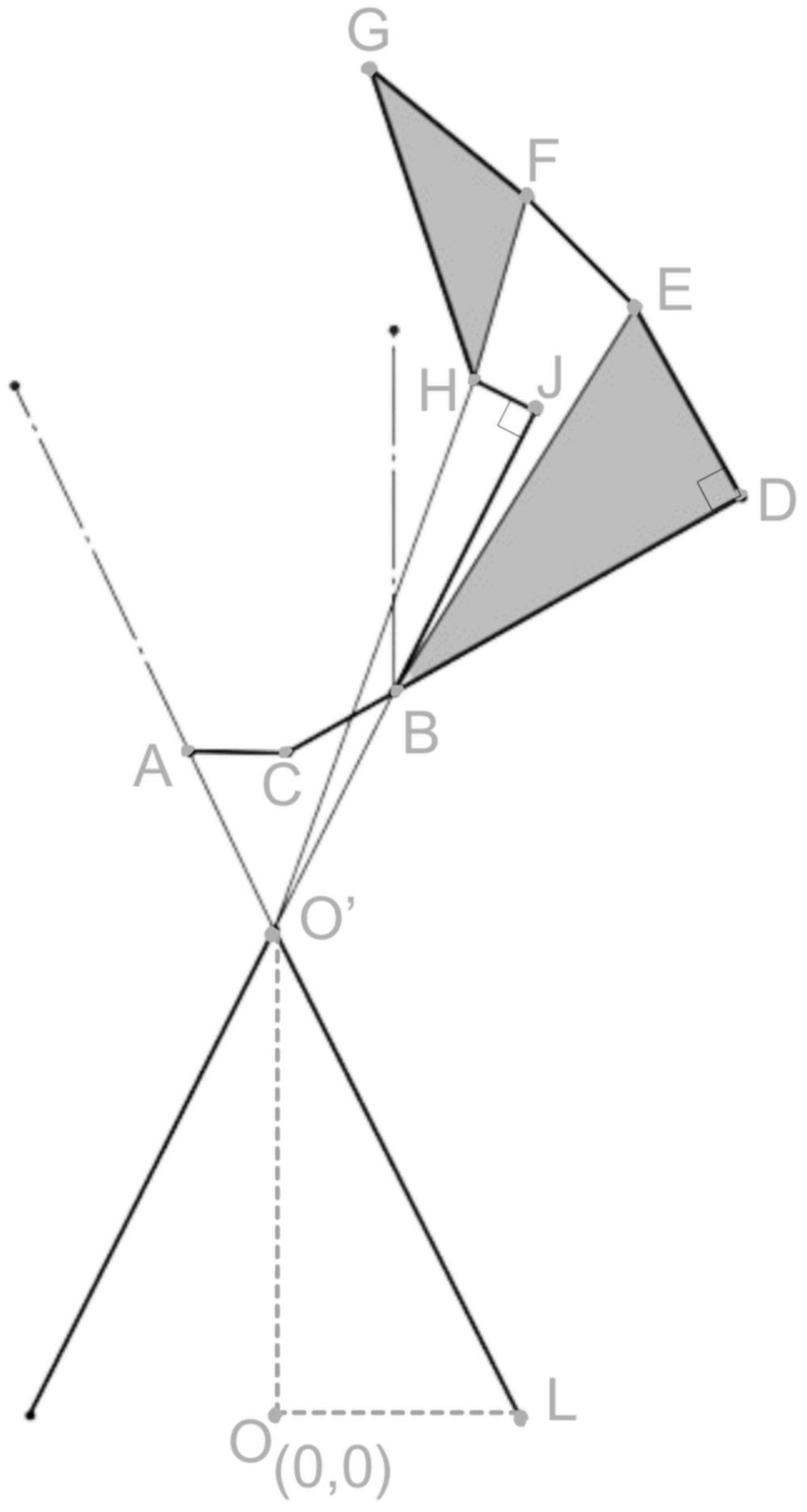


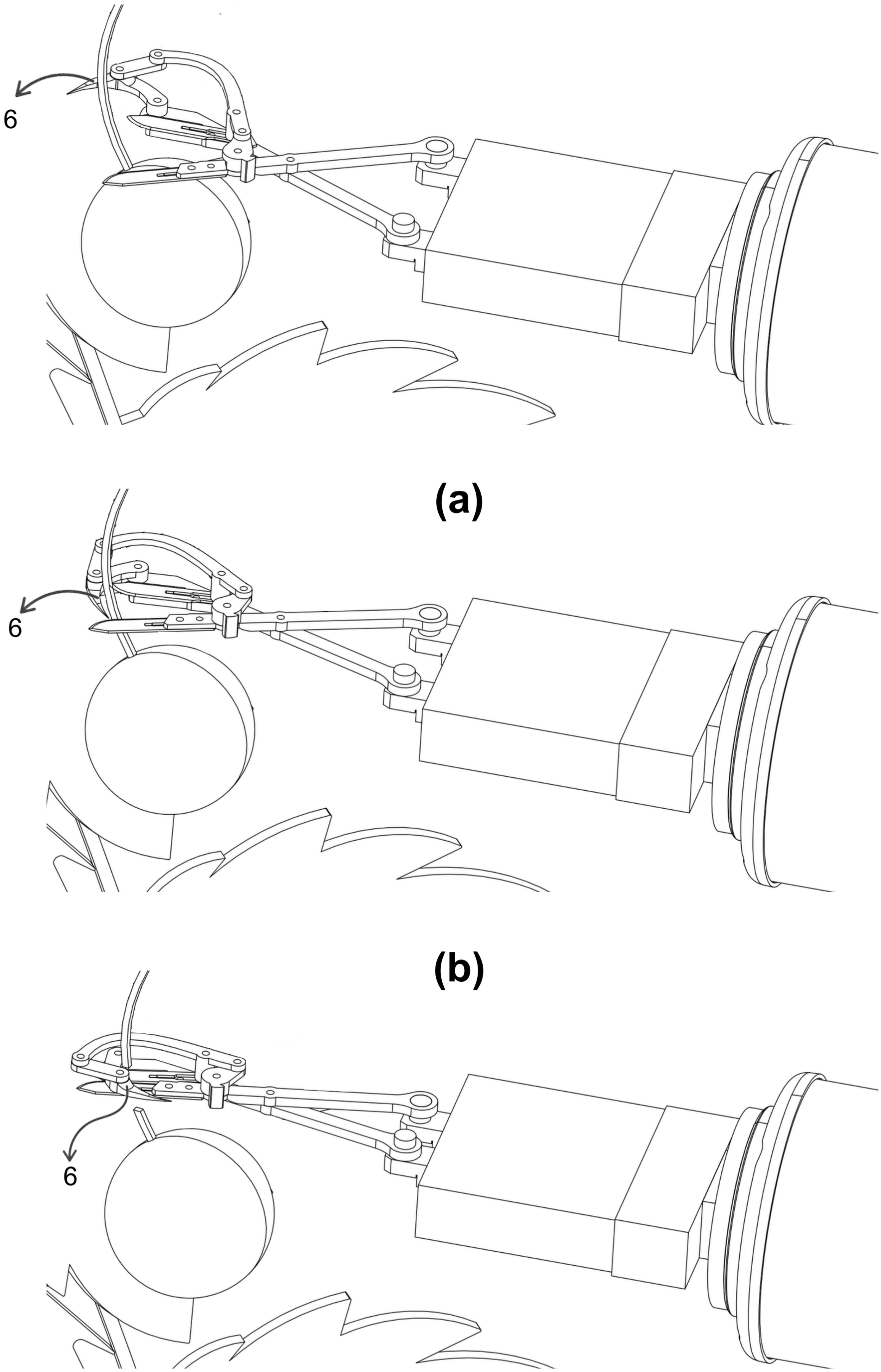
Fig 3.



(a)

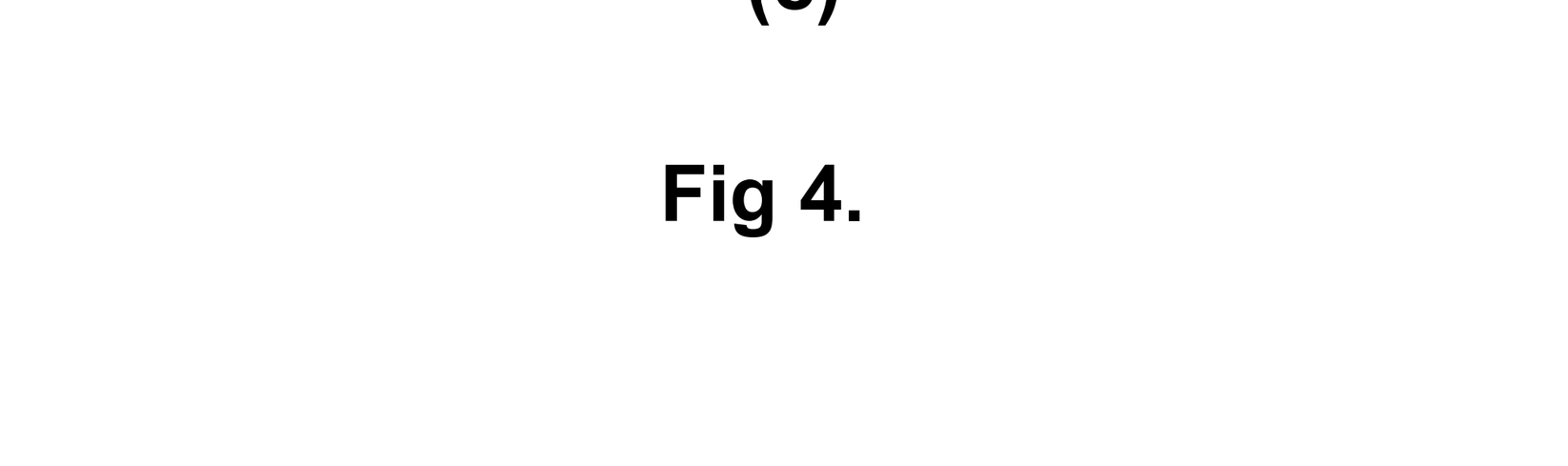


(b)



(a)

(b)



(c)

Fig 4.

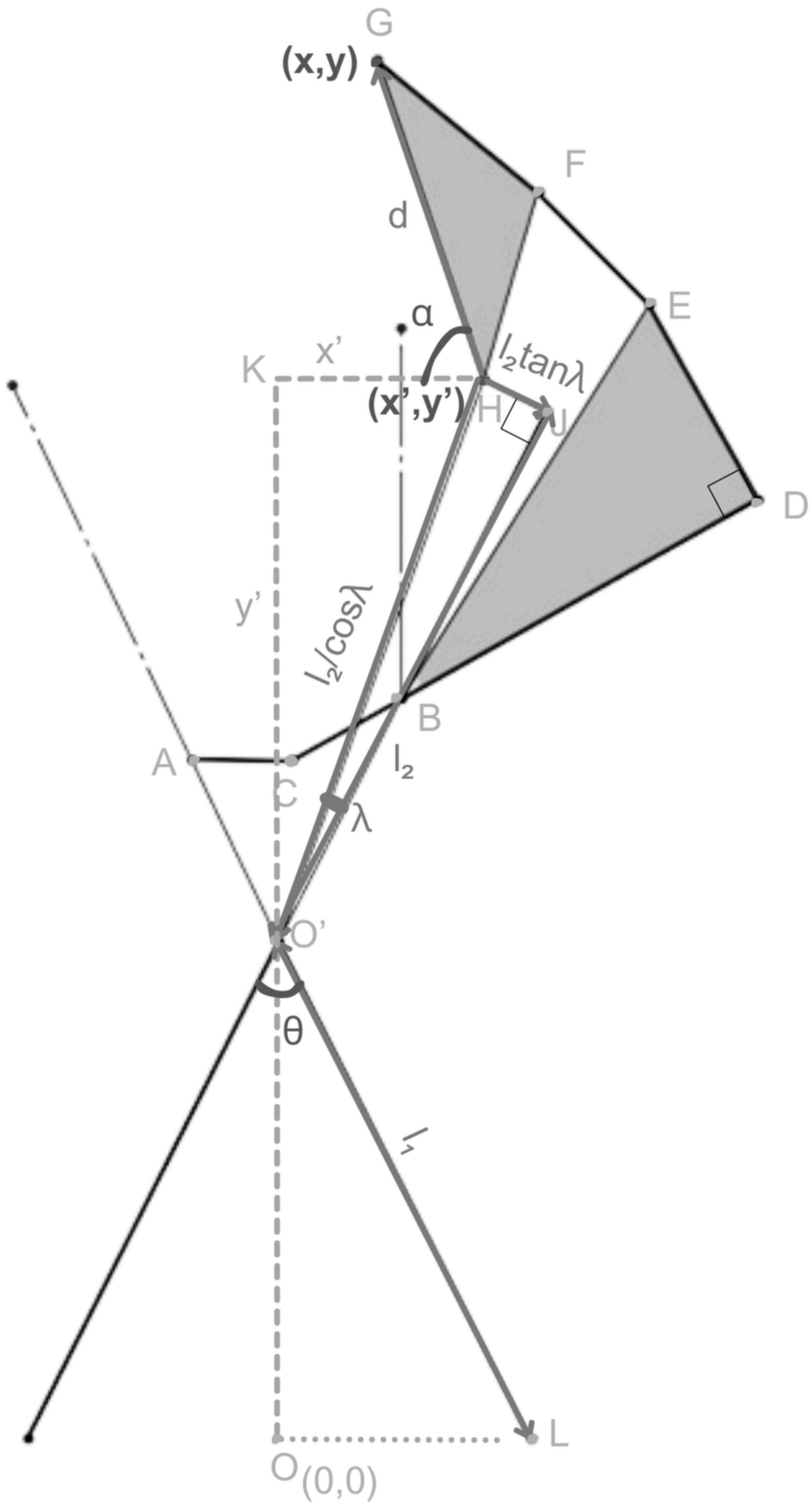
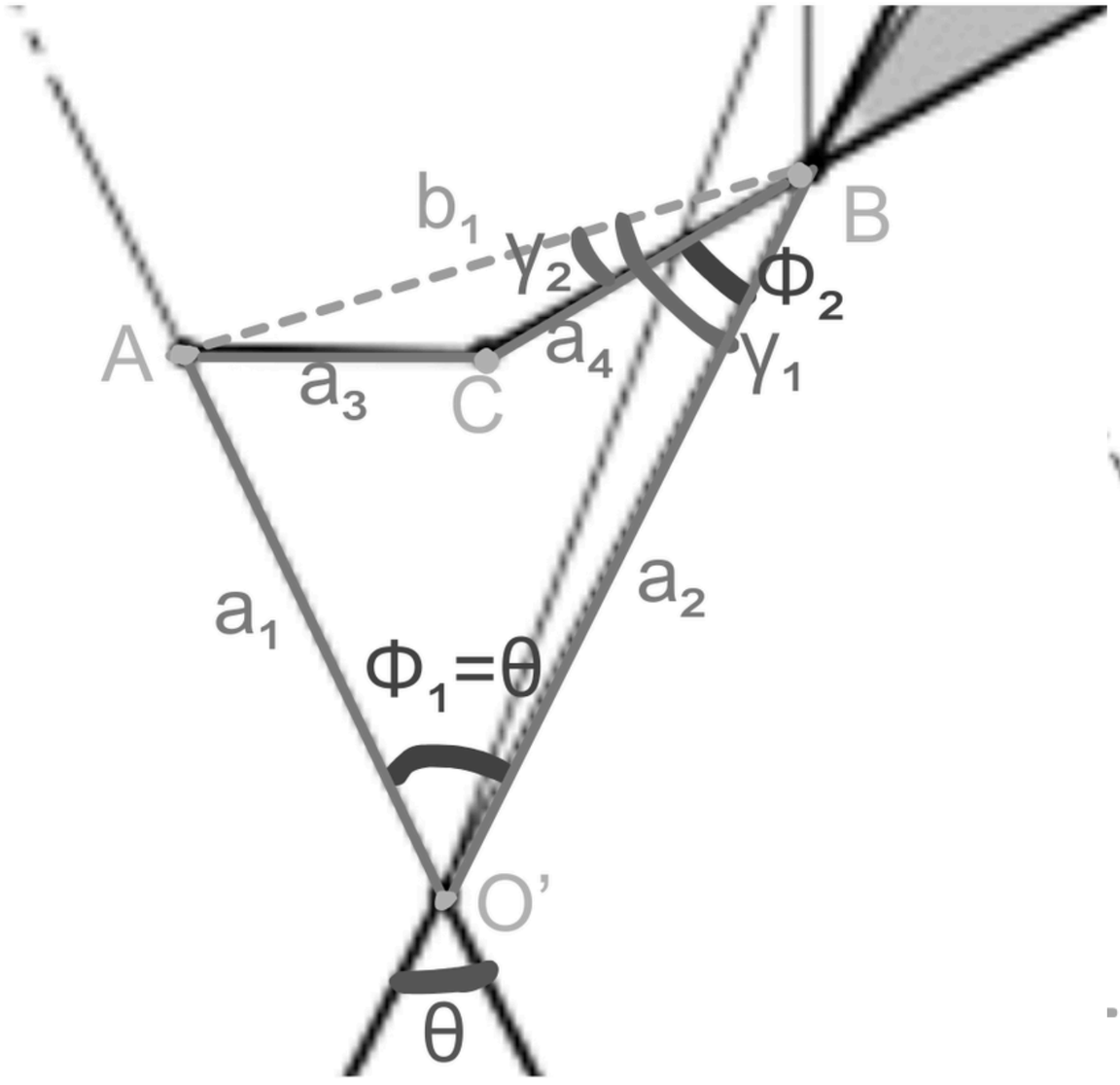
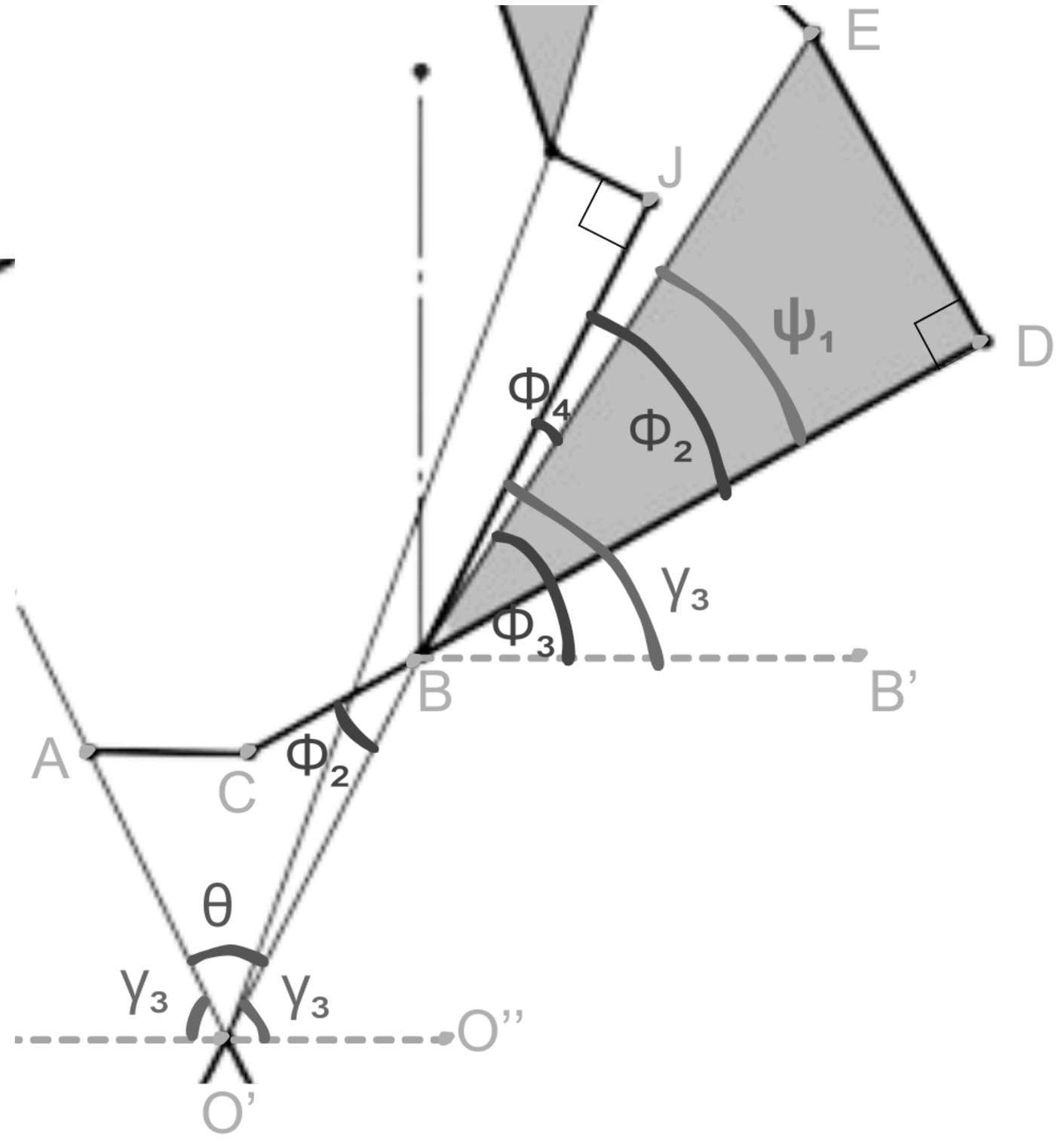

G
(x,y)
F
d
E
α
x'
K
$l_2 tan\lambda$
(x',y')
H
J
D
y'
$l_2/cos\lambda$
B
A
l_2
C
λ
O'
θ
l_1
L
O(0,0)

Fig 5.

Fig 6.


b1
γ2
Φ2
B
A
a3
C
a4
γ1
a1
a2
Φ1=θ
O'
θ

(a)


E
J
ψ1
D
Φ4
Φ2
γ3
Φ3
B
B'
A
C
Φ2
θ
γ3
γ3
O''
O'

(b)

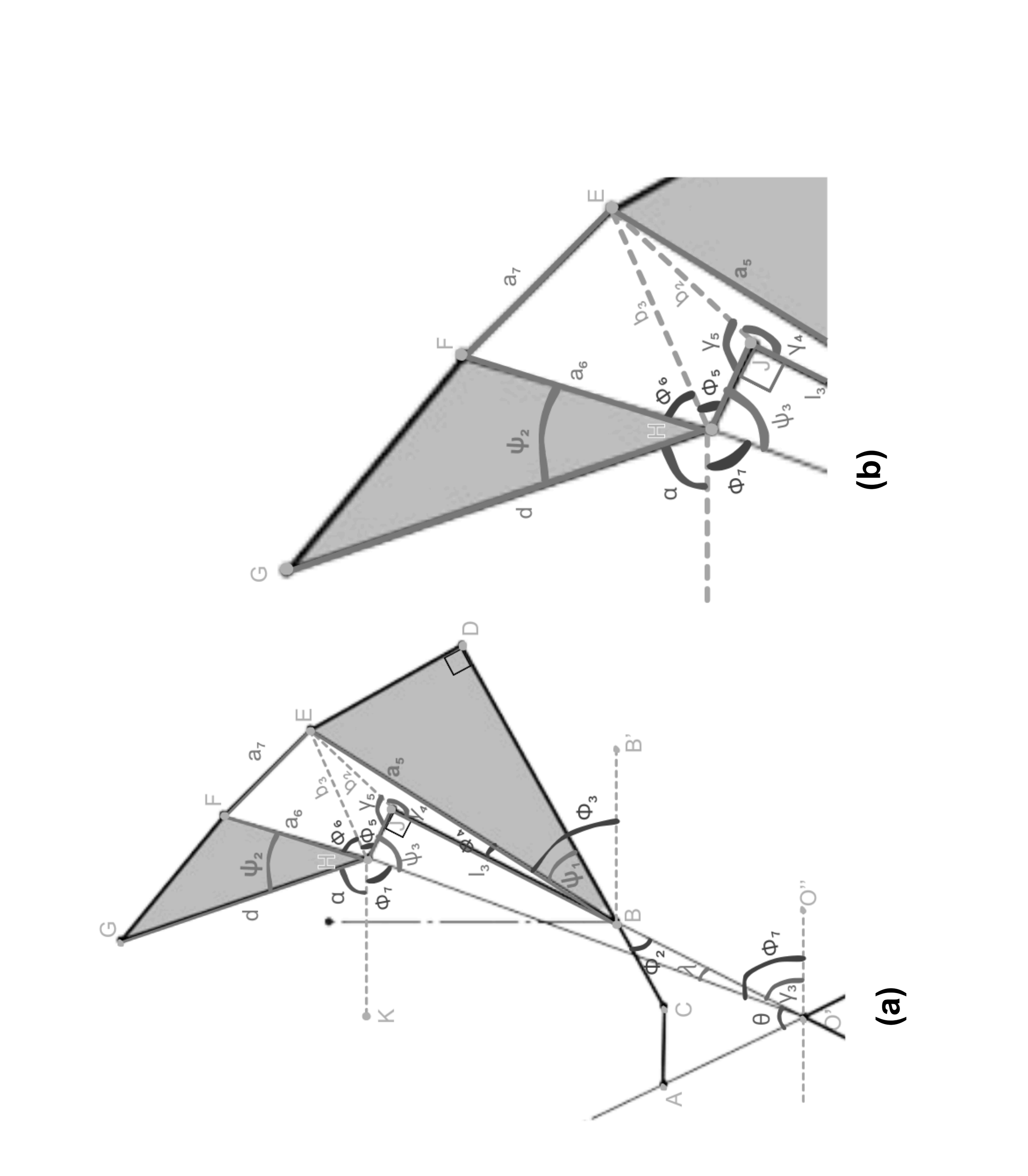
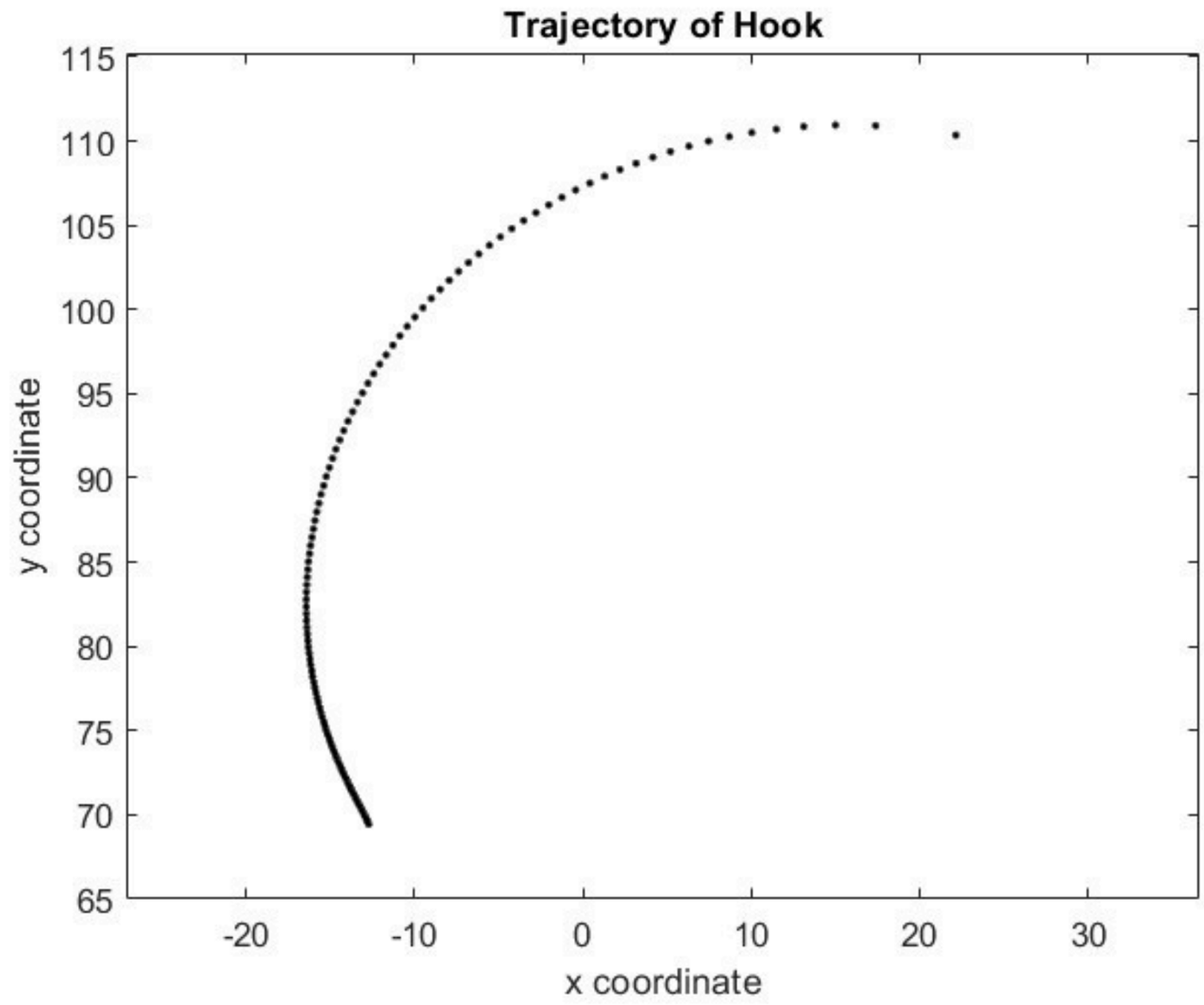
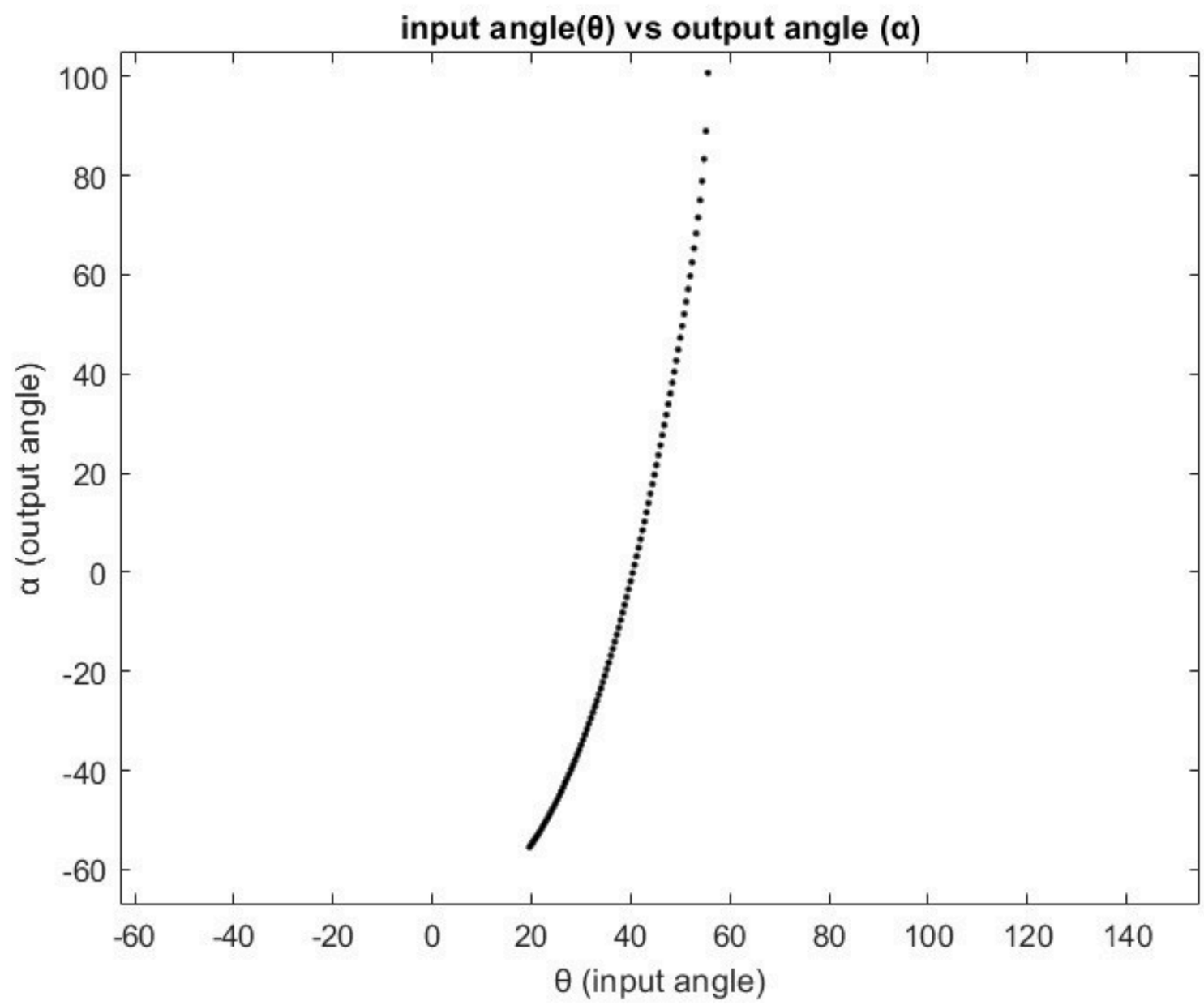

E
F
G
D
H
I
J
K
A
B
B'
C
O''
O'
a7
a5
a6
b3
b2
y5
y4
Φ6
Φ5
Φ4
Φ3
Φ2
Φ7
ψ3
ψ2
ψ1
l3
d
α
θ
γ3
(b)
(a)

Fig 7.


Trajectory of Hook
y coordinate
x coordinate
115
110
105
100
95
90
85
80
75
70
65
-20
-10
0
10
20
30

(a)


input angle(θ) vs output angle (α)
α (output angle)
θ (input angle)
100
80
60
40
20
0
-20
-40
-60
-60
-40
-20
0
20
40
60
80
100
120
140

(b)

Fig 10.

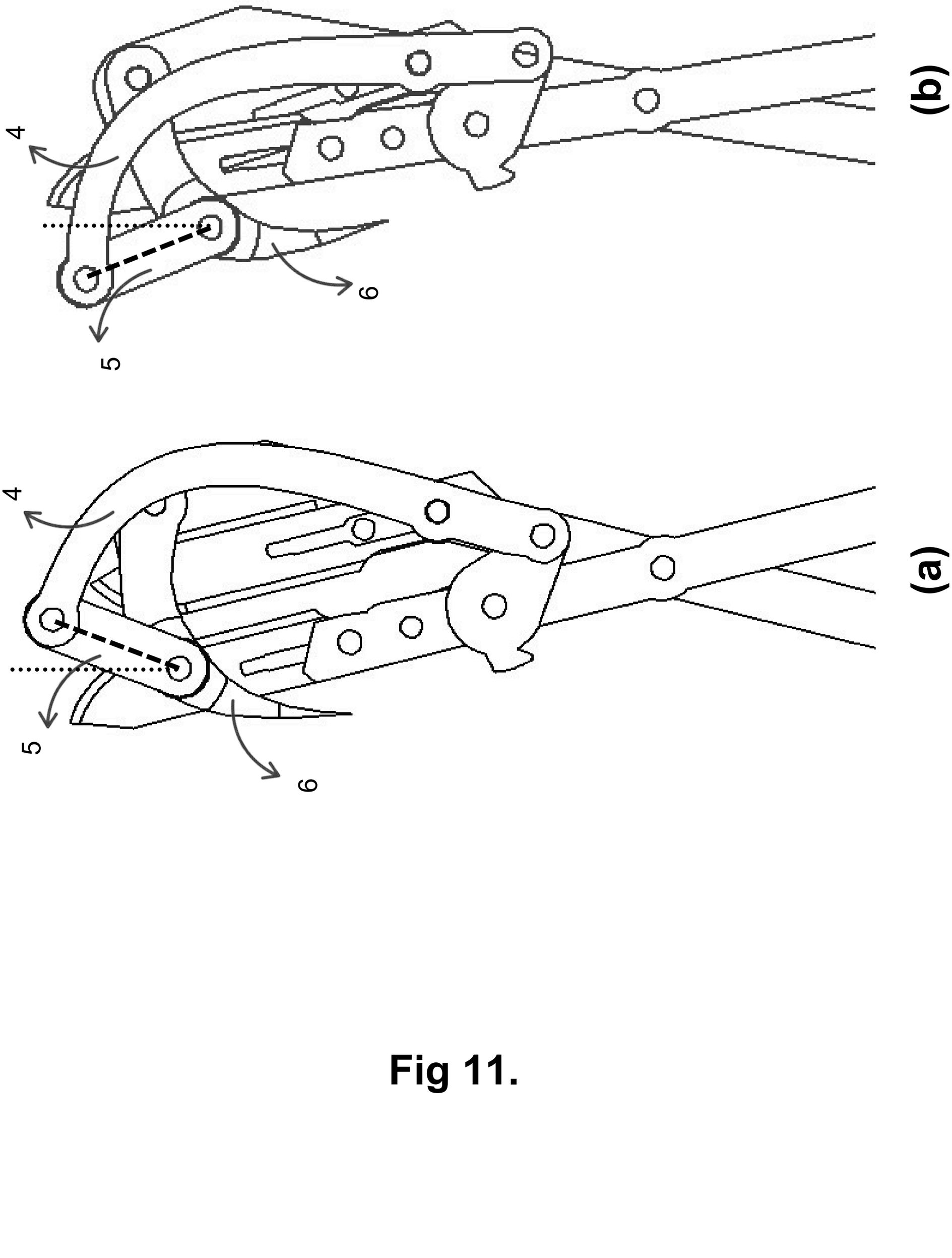


Fig 11.

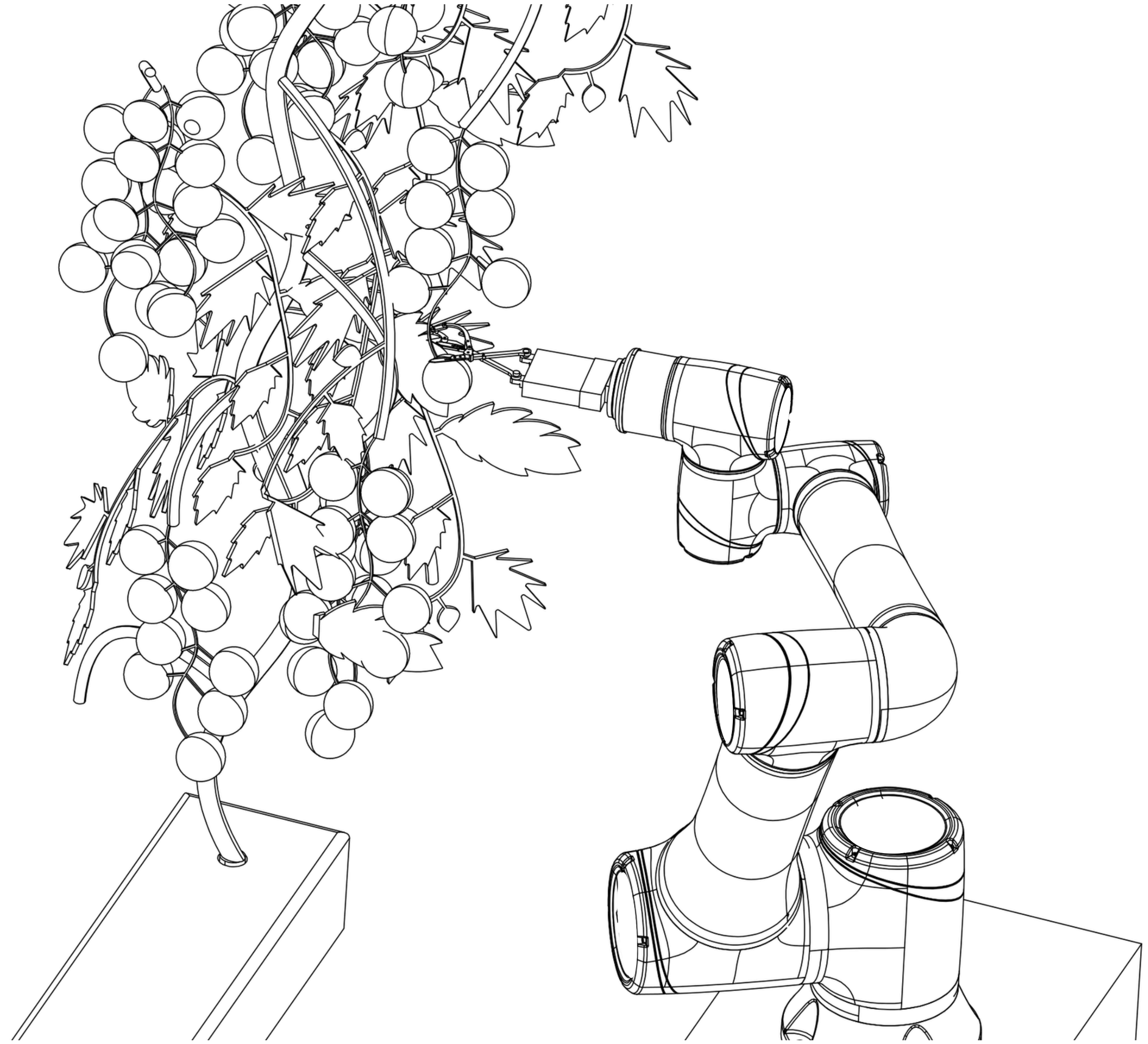
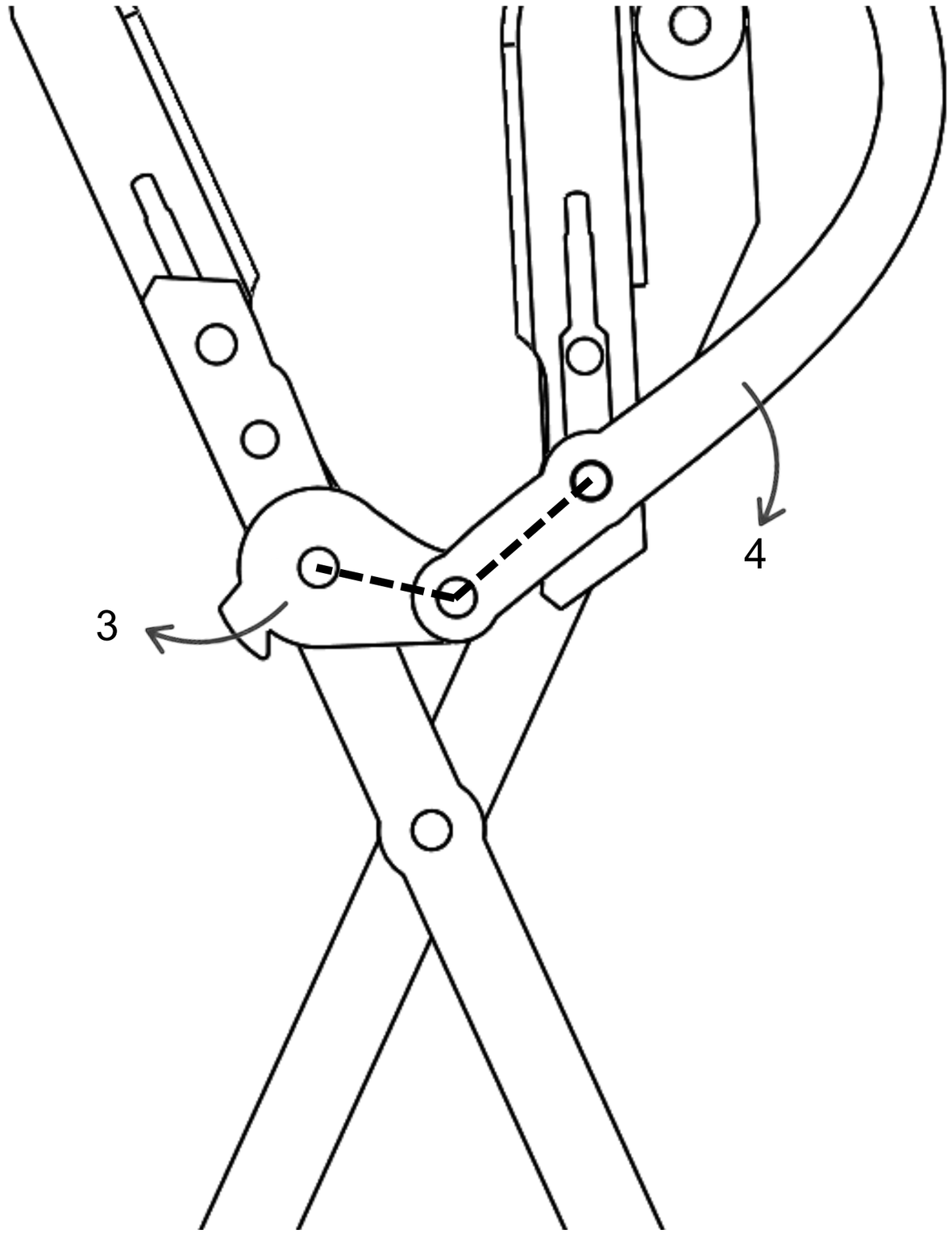
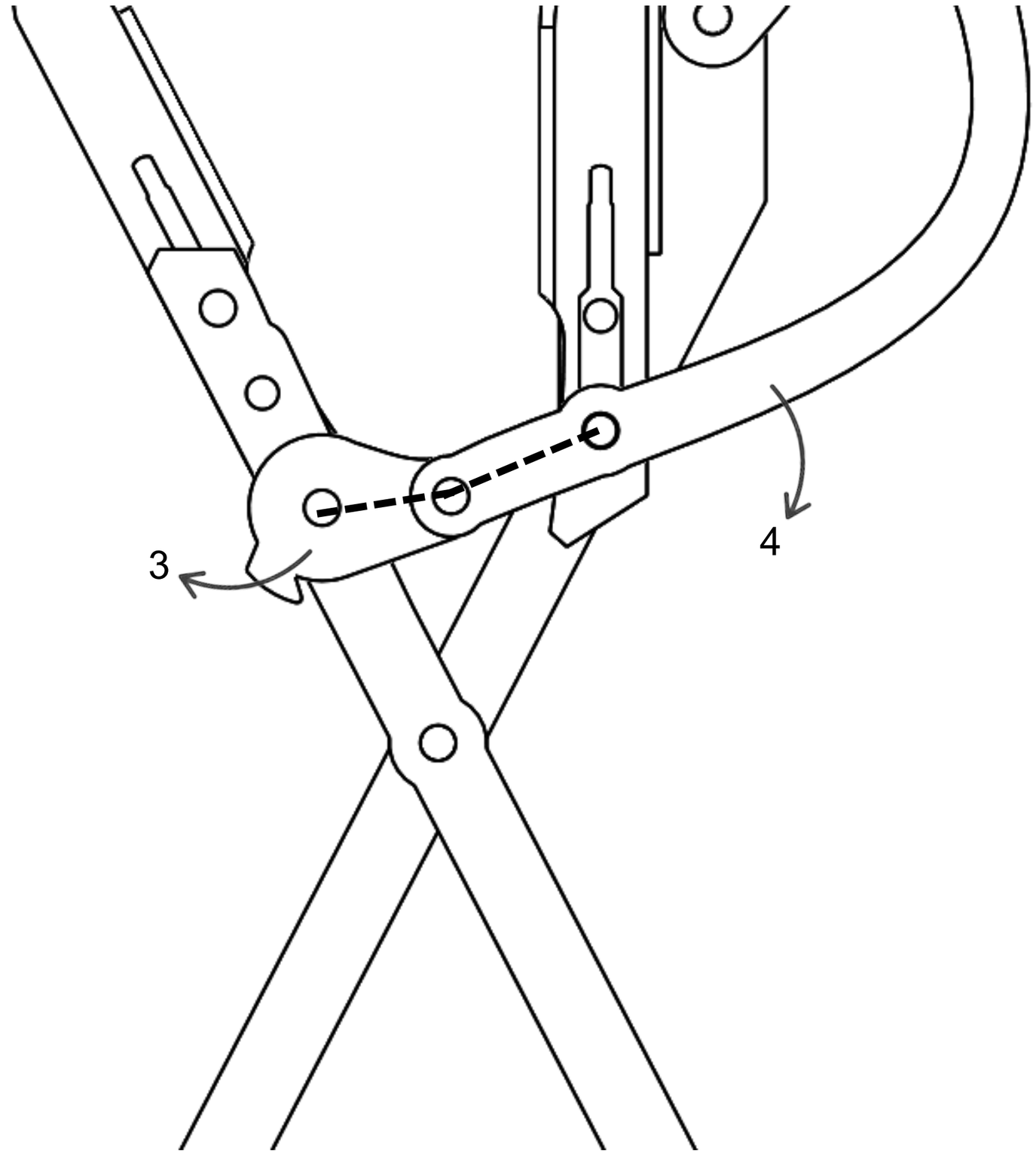


Fig 12.

Fig 13.



(a)



(b)